

CRAMBRUNCH

Submitted by:

SWAGATIKA PATI

Registration Number : **1292/23**

**Under The Guidance Of
Dr. Stitapragyan Mohanty**



Session: 2025-26

5 Years Integrated MCA

PG Department of Computer Science & Applications

Utkal University

PROJECT REPORT ON

CRAMBRUNCH

SUBMITTED IN PARTIAL FULFILLMENT FOR AWARD OF

DEGREE

IN

Integrated MCA

(BATCH 2023-2028)



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DECLARATION

I hereby declare that the project entitled “**CRAMBRUNCH**” submitted to the 5 year Integrated MCA, Utkal University, Bhubaneswar, Odisha in partial fulfillment for the award of the BCA degree in Integrated MCA for the session 2023-2026 is an authentic record of my own work carried out under the guidance of “**Dr. Stitapragyan Mohanty**” and that the project has not previously formed the basis for the award of any other degree. The report has been prepared in Compliance with the guidelines specified by the university.

Place: Bhubaneswar

Date :

SWAGATIKA PATI

Regd. No: 1292/23

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

GUIDE

Dr. STITAPRAGYAN MOHANTY

Lecturer

Dept.of Integrated MCA

ACKNOWLEDGMENT

I am very grateful to my project guide **Dr. STITAPRAGYAN MOHANTY** for giving her valuable time and constructive guidance in preparing the Synopsis/Project. It would not have been possible to complete this project in such a short period of time without her kind encouragement and valuable guidance.

SWAGATIKA PATI

Signature

Date:

CERTIFICATE

This is to certify that this project entitled “**CRAMBRUNCH**” submitted in partial fulfillment of the BCA degree in integrated MCA to the 5 year Integrated MCA, Utkal University, Bhubaneswar, Odisha, done by **Miss. SWAGATIKA PATI**, Registration No. 1292/23, is an authentic work carried out by her at Integrated MCA, Utkal University under guidance and is worthy of acceptance for the regulations of the university and in our opinion, it has reached the standard needed for submission.

Examiner

Dr. Stitapragyan Mohanty

GUIDE

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ABSTRACT

Introduction:-

CRAMBRUNCH is a smart, web-based platform developed to organize and manage academic and campus activities digitally. The main aim of this project is to reduce manual work and paperwork by storing all student and campus information in a single secure system. It provides role-based access for students, teachers, parents, and administrators to ensure proper data management, transparency, and controlled access.

The platform manages important academic activities such as maintaining student details, tracking attendance, monitoring academic performance, and handling library records. Attendance is calculated automatically, and a 75% attendance prediction feature helps students understand their eligibility to sit in the semester exams in advance. Along with academics, CRAMBRUNCH includes features like skill exchange between students, anonymous complaint submission with support-based escalation, teacher feedback, online class representative selection for more transparency and unbiased result, also automatically notifies parents via when a student's attendance falls below 75% and when academic results are declared and organized sharing of study materials, which improves overall campus involvement.

CRAMBRUNCH is developed using HTML5, CSS3, and JavaScript for the frontend, while Node.js and Express.js are used for backend development. MongoDB is used for storing data, along with tools like Chart.js for visual representation and Nodemailer for sending system emails. The project can be implemented in schools, colleges, and universities to improve academic management, communication, and administrative efficiency.

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CHAPTER-1

(INTRODUCTION)

1. INTRODUCTION:

In today's schools and colleges, managing students' information is very important but often difficult with old paper-based methods. CramBrunch is a simple, user-friendly web app that makes everything easier by bringing all student data and activities into one place for better organization and communication.

1.1. PROBLEM DEFINITION AND OBJECTIVE:-

❖ Current Challenges in Student Management:

- Schools and colleges rely on paper registers and manual records for attendance, grades, and feedback.
- This leads to frequent errors like lost files, miscalculated marks, and inaccurate attendance counts.
- No real-time updates mean parents are unaware of issues like low attendance or poor performance.
- Students hesitate to share honest feedback due to lack of anonymity, causing unresolved complaints.
- No platform for skill-sharing, so students miss opportunities to learn from peers (e.g., coding or music).

❖ Key Problems Identified:

- Time wasted on paperwork reduces teaching and learning time for staff and students.
- Poor communication between teachers, parents, and admins creates gaps in student support.
- Fragmented data systems make it hard to track academic progress or predict issues like dropouts.
- Scalability issues in growing institutions with hundreds of students.
- Lack of modern tools like alerts and exchanges limits student engagement and holistic development.

❖ **Main Objective of CramBrunch:**

- Build a single web-based app to manage all student data digitally and securely.
- Integrate academics, attendance, parent alerts, anonymous feedback, and skill exchange seamlessly.
- Automate routine tasks to save time and reduce errors by 80-90%.

❖ **Specific Goals:**

- Enable real-time attendance tracking with predictions to flag at-risk students early.
- Send instant SMS/email alerts to parents for low attendance or grades below threshold.
- Provide safe anonymous feedback system for complaints and suggestions.
- Create skill exchange marketplace for peer-to-peer learning (e.g., "teach guitar, learn math").
- Ensure role-based access: students view data, teachers update records, parents get reports, admins oversee all.
- Improve overall efficiency, transparency, and student outcomes in educational institutions.

1.2 Motivation:

❖ **Growing Administrative Burdens:**

- Schools and colleges face increasing student numbers, but staff still use outdated paper systems.
- Teachers spend hours on manual attendance, grading, and record-keeping instead of focusing on teaching.
- This leads to burnout, errors, and less time for personalized student support.

❖ **Poor Parent Involvement:**

- Parents rarely get timely updates on attendance drops or low grades, missing chances to help early.
- Without alerts, issues like absenteeism or academic struggles go unnoticed until exams or parent-teacher meetings.
- Strong parent-school communication improves student performance by 20-30%, but current methods fail here.

❖ **Student Engagement Gaps:**

- Students lack safe ways to give honest feedback or report bullying/complaints anonymously.
- No platforms exist for skill-sharing, so talented peers can't exchange knowledge (e.g., sports coaching for homework help).
- Modern students expect digital tools like apps for activities, events, and progress tracking.

❖ **Need for Data-Driven Decisions:**

- Manual records make it impossible to predict trends like attendance patterns or at-risk students.
- Institutions need analytics to plan resources, events, and interventions effectively.
- Digital systems enable real-time reports, cutting paperwork by 70% and boosting efficiency.

Technology Adoption Push :

- Post-pandemic shift to online learning highlights the need for integrated platforms.
- Web-based apps are accessible on phones/laptops, fitting busy schedules for all users.
- CramBrunch is inspired by successful edtech tools but customized for comprehensive campus management.

Broader Educational Benefits :

- Aims to create transparent, efficient schools where everyone—students, teachers, parents, admins—collaborates better.
- Fosters holistic development through features like skill exchange and feedback loops.
- Prepares institutions for future scalability with secure, role-based access and predictive insights.

1.3 Project Overview:

CramBrunch is an easy-to-use online platform designed for schools. It covers:

- Student profiles with full academic history (marks, classes, activities).
- Real-time attendance marking and reports.
- Instant SMS/email alerts to parents for issues like missing classes.
- Anonymous feedback box for students/teachers to report problems safely.
- Skill exchange marketplace where students offer and request skills (e.g., "Teach me guitar" or "Help with math").
- Extra tools like complaint handling, event calendars, and attendance forecasts.

Everyone gets their own login: students see their data, teachers update records, parents view reports, and admins control everything. It's secure, fast, and works on any device with internet.

1.4 Software Specification:

Component	Specification
Operating System	Windows, macOS, Linux, Android, iOS
Front-End	HTML, CSS, JavaScript, Bootstrap
Back-End	Node.js, Express
Database	MongoDB
Code Editor	Visual Studio Code
Hosting Environment	Localhost for development and testing
Browser Support	Latest versions of Chrome and Edge
Internet Requirement	Minimum 10 Mbps
Application Type	Web-based application
User Interface	Responsive and mobile-friendly
Security	Role-based access control, secure login
Data Storage	NoSQL database storage using MongoDB
Maintenance	Easy to update and manage
Scalability	Can be expanded for multiple colleges in future

These software specifications show the tools and environment required to run and develop the CRAMBRUNCH system smoothly. They also make the platform easy to use, maintain, and expand.

1.5 Hardware Specification:

Component	Minimum Requirement	Recommended Requirement
Processor	Any modern processor	Intel i3 / i5 or AMD Ryzen 3 / 5 or higher
RAM	4 GB	8 GB or more
Storage	16 GB free space	32 GB or more
Internet Speed	10 Mbps	20 Mbps or higher
Device Type	Desktop / Laptop / Tablet / Smartphone	Desktop / Laptop with good performance
Display	Standard screen resolution	HD display or above
Input Devices	Keyboard and Mouse	Keyboard, Mouse, and Touch support
Power Supply	Normal power backup	UPS or battery backup for uninterrupted use

1.6 Optimization of the Report:

Aspect	Optimization in CRAMBRUNCH Report
Data Collection	Collects data from all modules.
Accuracy	Reduces human errors in report generation.
Speed	Generates reports quickly in real time.
Data Storage	Maintains centralized digital records.
Accessibility	Provides role-based access to authorized users.
Attendance Reports	Generates attendance summaries .
Academic Reports	Displays marks and performance details efficiently.
Grievance Reports	Tracks complaints and resolution status.
Transparency	Improves visibility of institutional activities.
Paperwork Reduction	Minimizes manual documentation work.
Decision Making	Helps administrators make faster decisions.
Time Efficiency	Saves time for teachers and management.

CHAPTER-2

(Literature Survey)

2.Literature Survey:

Literature surveys on educational management systems reveal key challenges and opportunities in modern campuses. Traditional and partial digital approaches fall short in automation, accessibility, and integration.

❖ Manual Paper-Based Systems:

Many colleges rely on paper records for attendance, complaints, and student files. These methods demand high manual effort, leading to time loss during peak periods like exams or events. Common issues include lost documents, handwriting errors, and delays in sharing updates with parents.

❖ Problems with Traditional Methods:

Studies highlight frequent errors from manual data entry, such as mismatched attendance counts or unresolved complaints. Slow processing affects decision-making, like spotting at-risk students early. Poor communication creates gaps—teachers note absences on sheets, but parents learn late, reducing involvement.

❖ Existing Digital Solutions:

ERP systems like Fedena or custom software handle basics such as fees and grades. They offer centralized data but face limits in mobile access and real-time alerts. Most lack student-focused tools, like skill exchange or anonymous feedback portals.

❖ Gaps in Current Tools:

Commercial ERPs often ignore skill-building features or fast grievance resolution. They provide admin access but rarely role-based views for students, parents, or HODs. Scalability issues arise in multi-department colleges without strong notification engines.

❖ Research Recommendations

Recent papers stress web-based platforms for full automation. Role-based access ensures security—students see profiles, teachers mark attendance instantly. Integrated prediction tools, like attendance eligibility forecasts, boost efficiency. Surveys call for systems combining management with community growth, such as skill hubs, to prepare students for jobs.

2.1 Existing System:

Current school and college systems manage basic tasks like attendance, grades, and fees. ERP tools like Fedena or EduSys store student data centrally and handle simple library functions.

❖ Key Limitations:

- They miss modern features such as skill exchange between students.
- Feedback tools are just basic forms, not interactive apps.
- No real-time notifications for parents or teachers about absences or issues.
- Limited role-based access—admins see everything, but students get few personalized views.
- Manual processes still needed for complaints or leave requests, causing delays.
- Poor integration—no links between attendance, skills, or internships in one dashboard.
- High costs for setup and maintenance, not ideal for small colleges.

2.2 Proposed System :

CRAMBRUNCH is a new web-based platform to manage college activities fully online. It fixes issues in old systems by adding smart features, role-based access, and real-time updates for students, teachers, parents, and admins.

❖ Core Improvements:

- Automates attendance with one-tap marking and predicts eligibility to avoid shortages.
- Adds skill exchange where students trade skills like coding for design tips, with badges for proof.
- Handles complaints via anonymous tickets that auto-escalate if not fixed in 24 hours.

❖ Role-Based Modules:

- **Students:** View profiles, download notes, raise grievances, and join micro-internships.
- **Teachers:** Mark attendance, claim substitute classes, enter grades, and track internships.
- **Parents:** See daily attendance feeds, get low-attendance warnings, and access fee docs.
- **Admins/HODs:** Generate reports, approve leaves, balance faculty loads, and audit logs.

❖ **Key Features:**

- Real-time notifications for absences, substitutions, and escalations.
- Mobile-friendly design works on phones, tablets, and desktops.
- Central dashboard syncs data across all roles instantly.
- Free, lightweight tech stack for easy setup in any college.

2.3 Study of the System:

The study of the system examines how CRAMBRUNCH works as a complete college management platform. It covers the main components, data flow, user interactions, and benefits over manual or basic digital systems.

❖ **System Overview:**

CRAMBRUNCH integrates academic and campus activities into one secure web platform. It supports role-based access for students, teachers, parents, HODs, and super admins, reducing paperwork while improving transparency and speed. Key areas include attendance tracking, skill exchange, grievance handling, and real-time notifications.

❖ **Main Components:**

- **User Modules:** Separate dashboards for each role—students check eligibility and notes, teachers mark attendance, parents view feeds.
- **Core Engines:** Handle timetable scheduling, data sync, and automated alerts to connect all modules instantly.
- **SkillHub:** Enables peer skill trading and micro-internships with badges for portfolios.
- **Admin Tools:** Audit logs, reports, and backups ensure security and performance.

❖ **Data Flow and Processes:**

Data moves smoothly across modules—for example, a teacher's attendance mark updates student profiles and triggers parent warnings in seconds. Grievances auto-escalate after 24 hours if unresolved. Predictive algorithms calculate safe lecture misses based on timetables.

❖ **User Interactions:**

Users log in to role-specific homepages. Simple actions like one-tap attendance or ticket submission work on any device. Notifications keep everyone updated via email or in-app alerts, making daily tasks quick and error-free.

❖ **Benefits and Feasibility:**

This setup cuts manual work by 80%, boosts communication, and scales for multiple departments. Built on free tools like Node.js and MongoDB, it's low-cost and easy to maintain, ideal for colleges seeking efficiency without complexity.

2.4 Feasibility Study:

The feasibility study checks if CRAMBRUNCH can be built, run, and scaled successfully. It looks at technical, economic, operational, and schedule aspects to confirm the project is practical for colleges.

❖ **Technical Feasibility:**

- CRAMBRUNCH uses proven web technologies like HTML, CSS, Node.js, Express, and MongoDB—all free and widely supported.
- Modern devices (phones, laptops) handle the lightweight app easily, with no need for special hardware.
- The stack supports real-time features like attendance sync and notifications without performance issues.

❖ **Economic Feasibility:**

- All tools are open-source and cost nothing to start or maintain—no licenses or cloud fees needed for localhost setup.
- It saves colleges money by cutting paper, printing, and staff time for manual tasks like attendance sheets.
- Long-term ROI comes from faster admin work and better student outcomes, paying back setup in months.

❖ **Operational Feasibility:**

- Users (students, teachers, parents) adapt quickly to simple interfaces like one-tap attendance or ticket forms.
- Role-based access matches daily college workflows—no big training required.
- Staff can manage it with basic IT skills, and mobile design fits busy campus life.

❖ **Schedule Feasibility:**

- Development timeline fits student projects: core modules ready in weeks using VS Code and existing code patterns.
- Testing covers units, integration, and user acceptance in short phases.
- Rollout to one department first, then scale—full system live in 2-3 months.

CHAPTER-3

(System Analysis and Design)

3. System Analysis and Design:

System analysis and design help shape the CRAMBRUNCH platform's structure, parts, and how they work together. This part reviews the current system, spots problems like manual tasks, no predictions, and poor visibility, and suggests a better web-based fix. It lists key features: handling attendance, complaints, skill sharing, and different access levels for students, teachers, parents, and admins. The design uses a flexible client-server setup with modern tech for real-time updates, strong security, and easy-to-use screens. In short, it creates a smart, automated campus system that's all connected.

❖ Requirement Specification:

Requirement specs list what the CRAMBRUNCH platform must do (functional needs) and how well it should do it (non-functional needs) to run smoothly and please users. Key features include login, role-based access, attendance tracking with predictions, complaint handling, skill sharing, and instant alerts for students, teachers, parents, and admins. It also covers security, growth potential, speed, reliability, and easy interfaces. Plus, it notes needed hardware and software for use on various devices. Overall, it gives a clear plan for a strong, safe campus system.

3.1 Functional Requirements :

Module	Functional Requirements
Authentication & Authorization	The system shall allow users to register and login using JWT authentication. The system shall support role-based access control for Student, Teacher, HOD, Parent, and Super Admin.
Student Management	The system shall allow admins and HODs to create, update, and manage student records.
Attendance Management	Teachers shall be able to mark attendance. Students and parents shall view attendance summaries. The system shall calculate attendance percentages and predict 75% eligibility.
Marks Management	Teachers shall enter internal, practical, assignment, and semester marks. Students and parents shall view academic performance.
Grievance Management	Students shall submit anonymous grievances. HOD/Admin shall update grievance status, respond, and escalate unresolved issues.
SkillHub Module	Students shall post and apply for skill barter or internship tasks. Admin shall award badges after task completion.
Study Materials	Teachers shall upload study materials using Google Drive links. Students shall download materials.

Notification System	The system shall send real-time notifications for attendance warnings, grievance updates, and announcements.
Timetable Management	HOD and teachers shall create and manage department timetables.
Feedback System	Students shall submit teacher feedback with star ratings. Teachers and HOD shall view aggregated feedback reports.
Parent Dashboard	Parents shall monitor ward attendance, marks, and receive alerts for low attendance.
Admin Management	Super Admin shall manage all users, grievances, study materials, and system health overview.
Reporting & Analytics	The system shall generate attendance reports, charts, and performance analytics using Chart.js.

3.2 Non-Functional Requirements :

Category	Non-Functional Requirements
Performance	The system should respond to user requests within 2–3 seconds under normal load.
Security	Passwords must be encrypted using bcryptjs. JWT authentication must secure all protected routes.
Scalability	The system should support increasing numbers of users and modules without performance degradation.
Availability	The platform should be available 24/7 with minimal downtime.
Reliability	Data such as attendance, marks, and grievances must be stored accurately without loss.
Usability	The interface should be simple, responsive, and easy to navigate for all user roles.
Maintainability	The project should follow modular architecture for easy maintenance and updates.
Compatibility	The application should work on modern browsers and devices.
Portability	The system should run on local servers as well as cloud platforms like MongoDB Atlas.
Data Integrity	Database operations should maintain consistency and prevent duplicate or invalid records.

Notification Efficiency	Notifications and alerts should be delivered in real time.
Backup & Recovery	Database backup and recovery mechanisms should be supported.

3.3 Database :

CRAMBRUNCH uses MongoDB as its main database. MongoDB is a NoSQL database that stores data in flexible document form, which is useful for a system with many modules and different types of records.

TABLE 3.1 – USER_SCHEMA:

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
Name	String	Not Null
Email	String	Unique, Not Null
password	String	Not Null
Role	String	Not Null
department	String	Not Null
semester	Number	Not Null
parentId	ObjectId	Foreign Key
isActive	Boolean	Default True
createdAt	Date	Not Null

TABLE 3.2 – ATTENDANCE_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
studentId	ObjectId	Foreign Key, Not Null
subject	String	Not Null
date	Date	Not Null
status	String	Not Null
markedBy	ObjectId	Foreign Key

TABLE 3.3 – MARKS_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
studentId	ObjectId	Foreign Key, Not Null
subject	String	Not Null
internal	Number	Not Null
practical	Number	Not Null
assignment	Number	Not Null
semesterExam	Number	Not Null
total	Number	Not Null

TABLE 3.4 – GRIEVANCE_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
studentId	ObjectId	Foreign Key
title	String	Not Null
description	String	Not Null
status	String	Default Pending
anonymous	Boolean	Default False
response	String	Null Allowed
createdAt	Date	Not Null

TABLE 3. 5 – MATERIAL_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
title	String	Not Null
subject	String	Not Null
driveLink	String	Not Null
uploadedBy	ObjectId	Foreign Key
uploadedAt	Date	Not Null

TABLE 3. 6 – SKILLHUB_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
title	String	Not Null
description	String	Not Null
postedBy	ObjectId	Foreign Key
applicants	Array	Not Null
status	String	Default Open
badgeAwarded	Boolean	Default False

TABLE 3. 7 – NOTIFICATION_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
userId	ObjectId	Foreign Key
message	String	Not Null
type	String	Not Null
isRead	Boolean	Default False
createdAt	Date	Not Null

TABLE 3.8 – TIMETABLE_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
department	String	Not Null
semester	Number	Not Null
subject	String	Not Null
faculty	String	Not Null
day	String	Not Null
time	String	Not Null

TABLE 3.9 – FEEDBACK_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
studentId	ObjectId	Foreign Key
teacherId	ObjectId	Foreign Key
rating	Number	Not Null
comment	String	Not Null
submittedAt	Date	Not Null

TABLE 3.10 – BADGE_SCHEMA

COLUMN NAME	TYPE	CONSTRAINT
_id	ObjectId	Primary Key
studentId	ObjectId	Foreign Key
badgeName	String	Not Null
description	String	Not Null
awardedAt	Date	Not Null

3.5 ER DIAGRAM:-

The ER diagram of the CRAMBRUNCH Smart Campus Management System represents the relationship between entities such as Users, Attendance, Marks, Grievances, Notifications, and Timetables. It helps in designing the database structure by defining how data is stored, connected, and managed within the system.

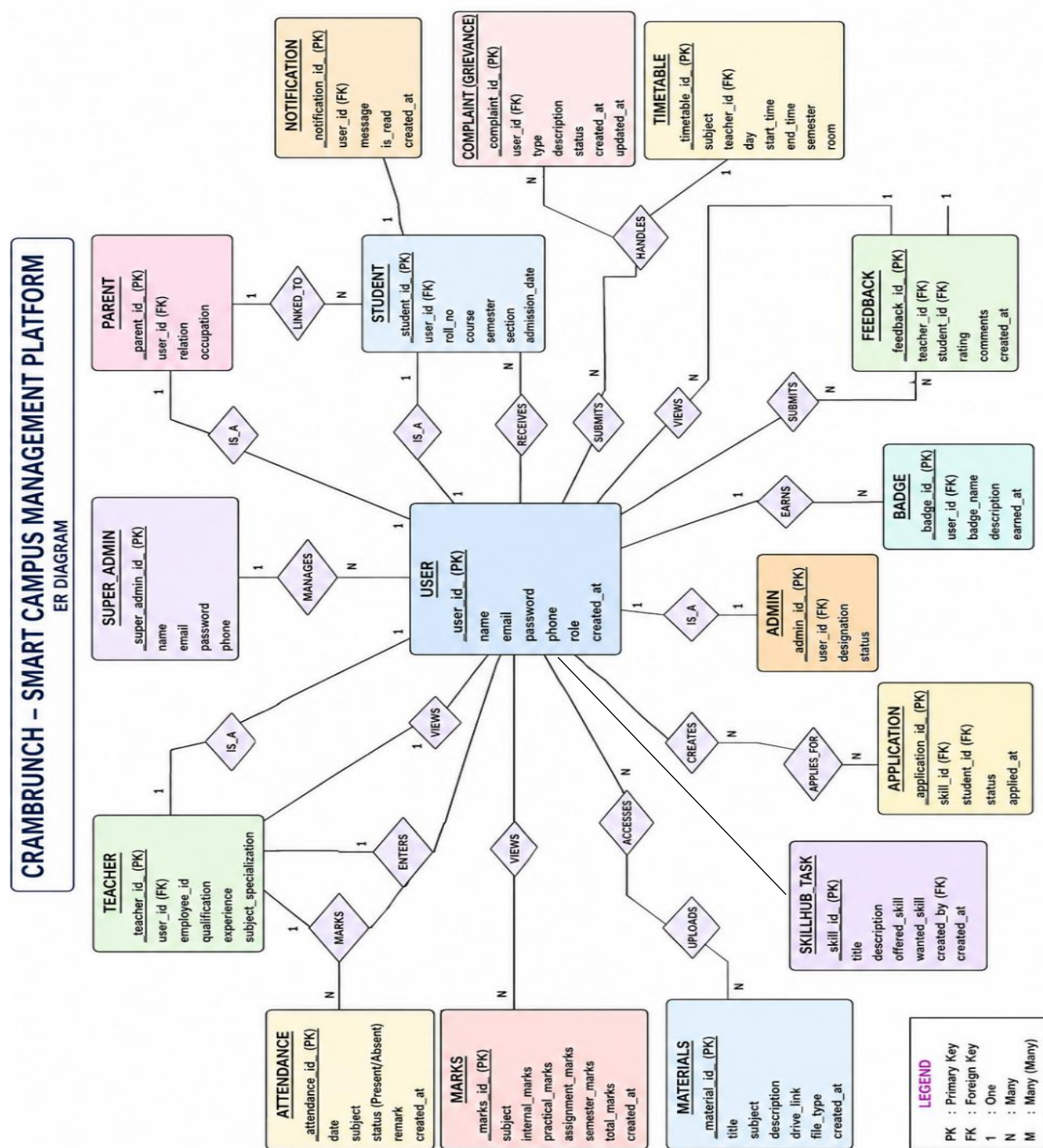
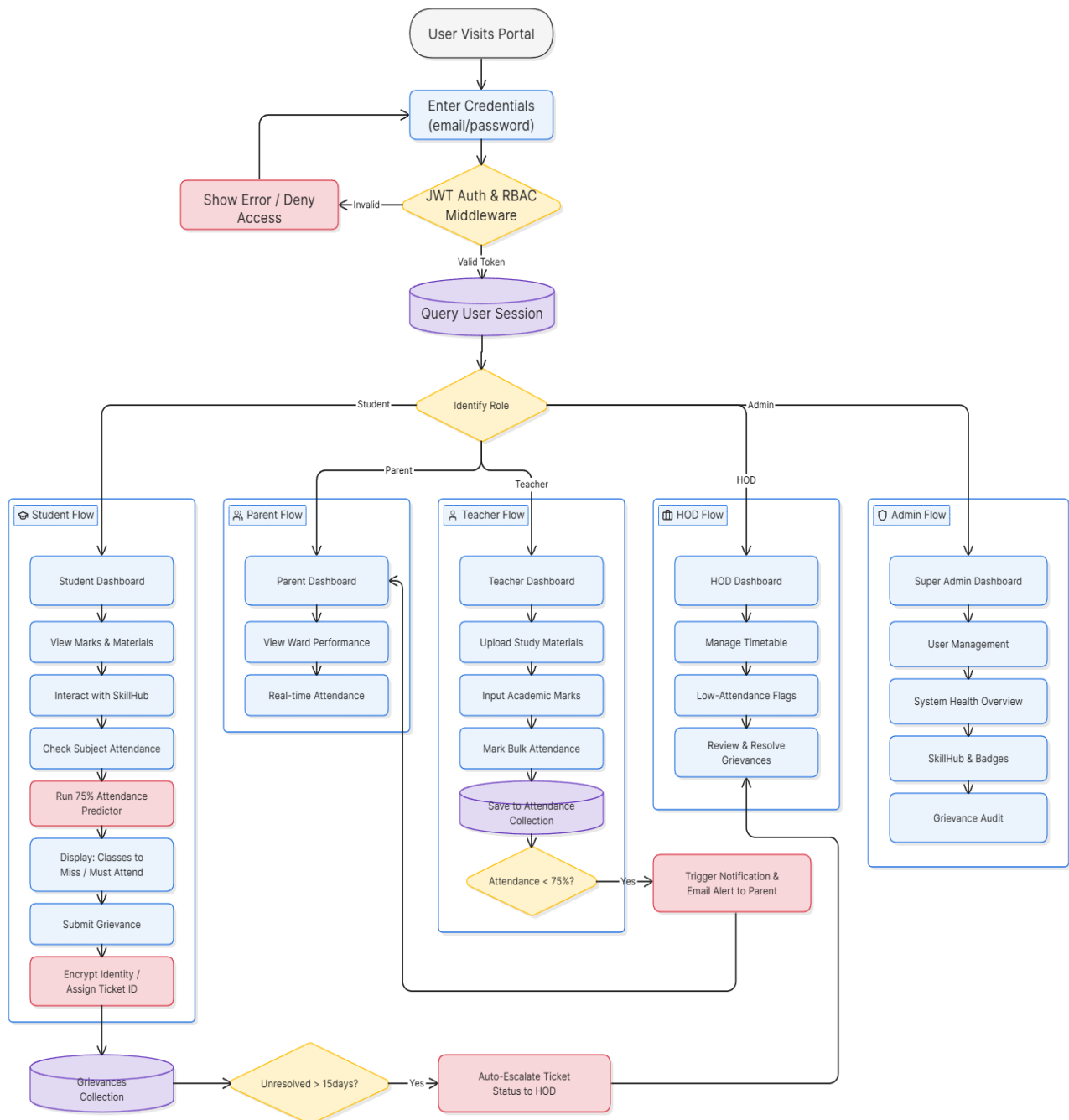


FIGURE -1 E-R DIAGRAM

3.4 Flow Charts

The flowchart represents the working process of the CRAMBRUNCH Smart Campus Management System, showing how different user roles interact with the platform through authentication and role-based modules. It illustrates the flow of data between student, teacher, parent, admin modules, the core engine, and the database management system for efficient campus operations.



[FIGURE 2. FLOW CHART

3.6 DATA FLOW DIAGRAM [DFD]:-

The Data Flow Diagram (DFD) of the CRAMBRUNCH Smart Campus Management System illustrates the flow of information between users, system modules, processes, and the database. It shows how data such as attendance, marks, grievances, notifications, and feedback are processed and shared among students, teachers, parents, HODs, and administrators.

3.6.1 DFD – LEVEL (0) :-

DFD – LEVEL 0 (CONTEXT DIAGRAM)

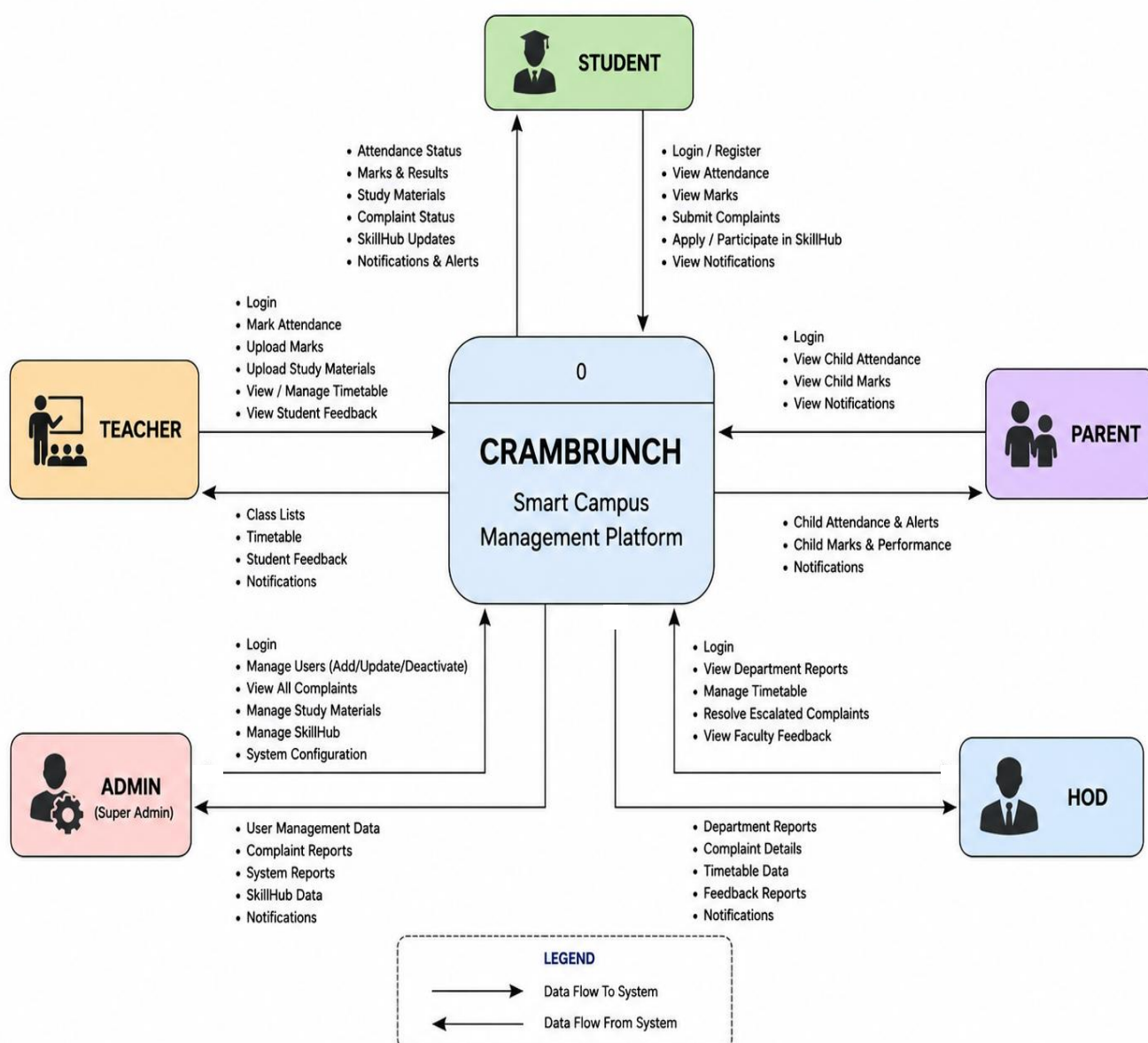


FIGURE -3 DATA FLOW DIAGRAM

3.6.2 DFD – LEVEL (1) :-

DFD – LEVEL 1 DIAGRAM CRAMBRUNCH – SMART CAMPUS MANAGEMENT PLATFORM

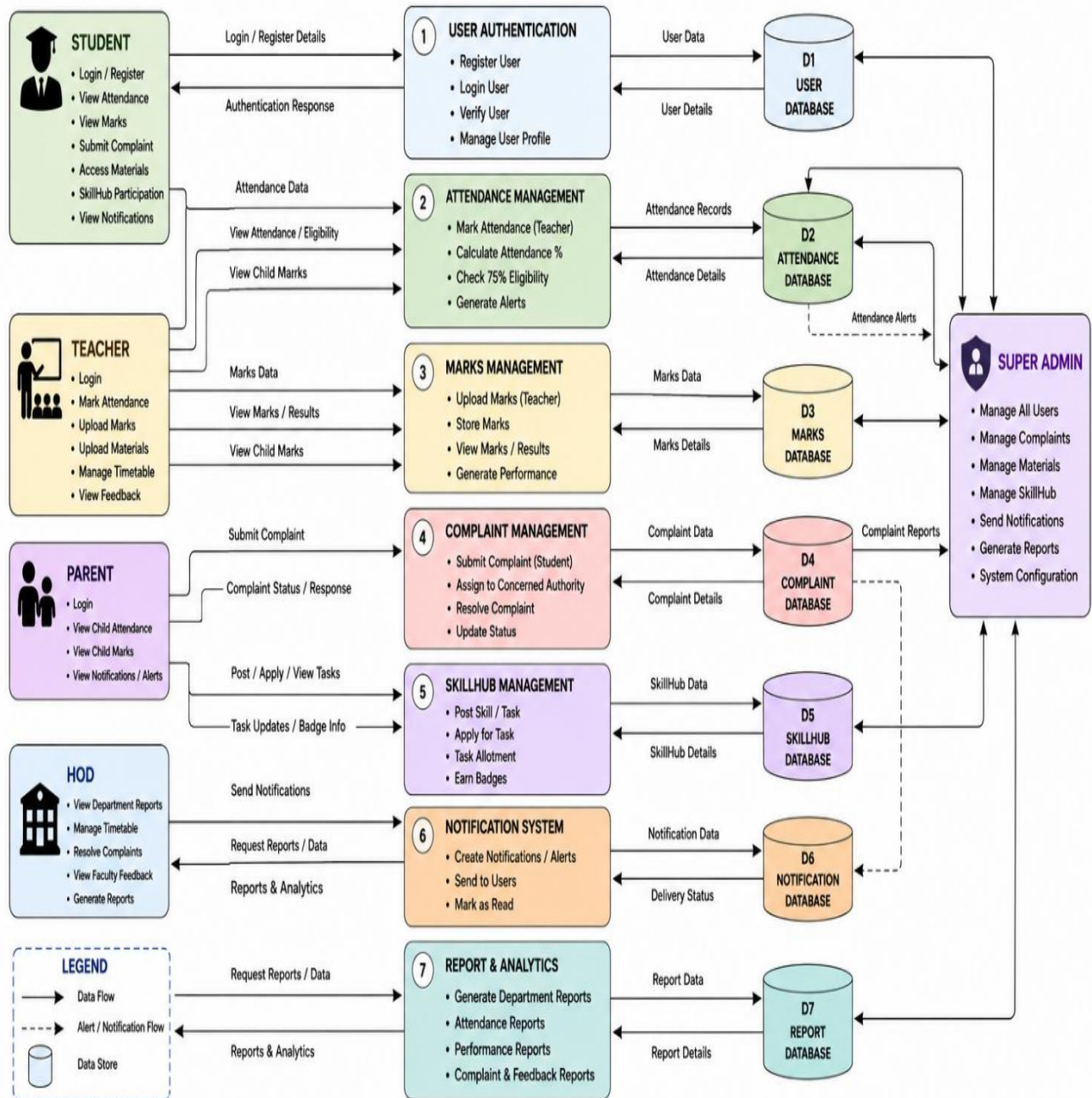


FIGURE -4 DATA FLOW DIAGRAM

3.7 UML DIAGRAM :-

The UML Diagram of the CRAMBRUNCH Smart Campus Management System represents the structure and interaction of different system components such as users, modules, classes, and processes. It helps in visualizing system design, relationships, functionalities, and workflow for better software development and maintenance.

3.7.1 USE CASE DIAGRAM:-

The Use Case Diagram of the CRAMBRUNCH Smart Campus Management System represents the interaction between different users and the system functionalities. It illustrates how students, teachers, parents, HODs, and administrators perform tasks such as attendance management, marks entry, grievance handling, notifications, and timetable management within the platform.

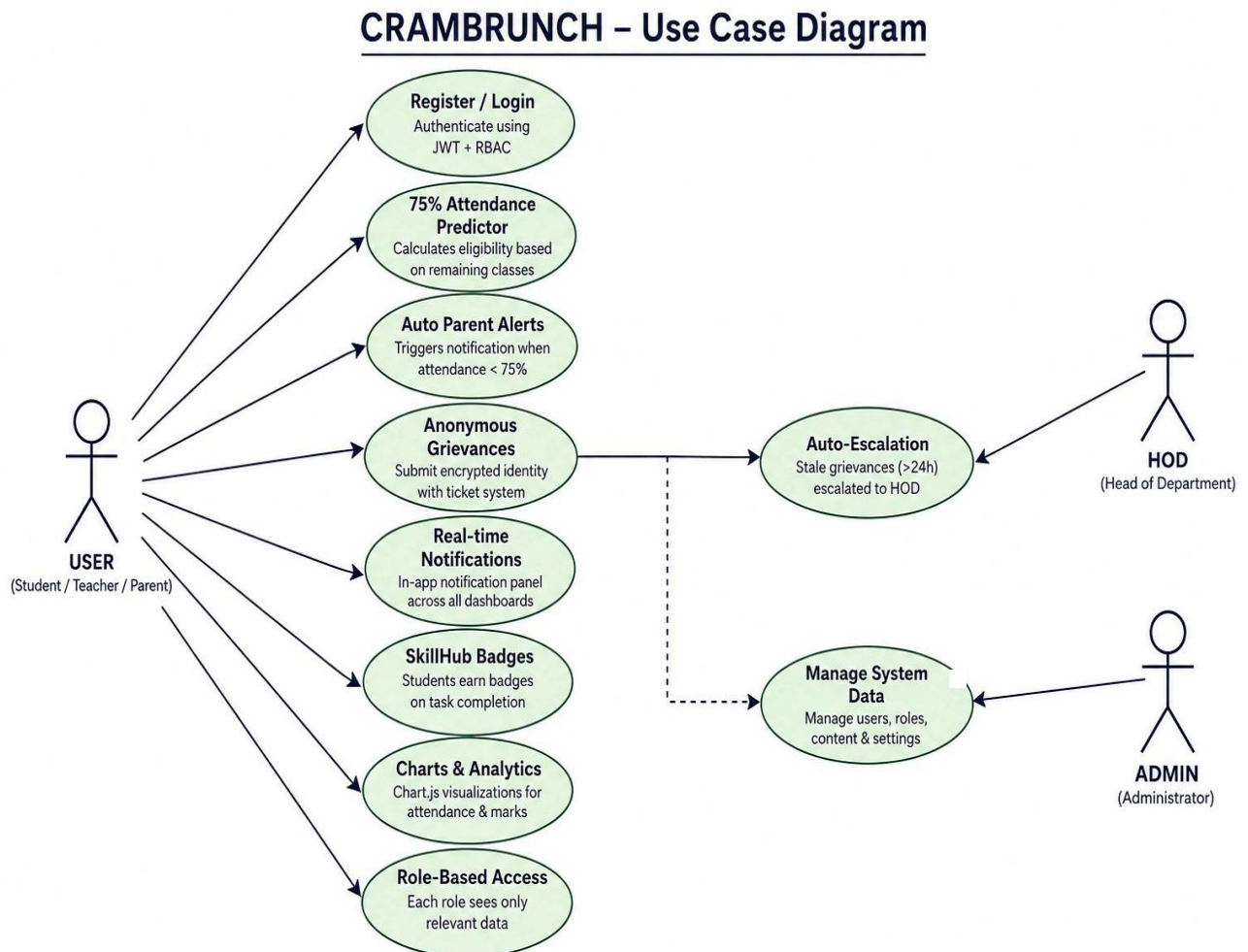


FIGURE -5 USE CASE DIAGRAM

3.7.2 ACTIVITY DIAGRAM:-

The Activity Diagram of the CRAMBRUNCH Smart Campus Management System represents the step-by-step workflow of system activities and user operations. It shows the sequence of processes such as login, attendance marking, marks management, grievance submission, notifications, and report generation within the platform.

3.7.2.1 ACTIVITY DIAGRAM [parent]:-

The Parent Activity Diagram of the CRAMBRUNCH Smart Campus Management System represents the sequence of activities performed by parents within the platform. It includes login, viewing the ward's attendance and academic performance, receiving low-attendance alerts, checking notifications, and monitoring overall student progress.

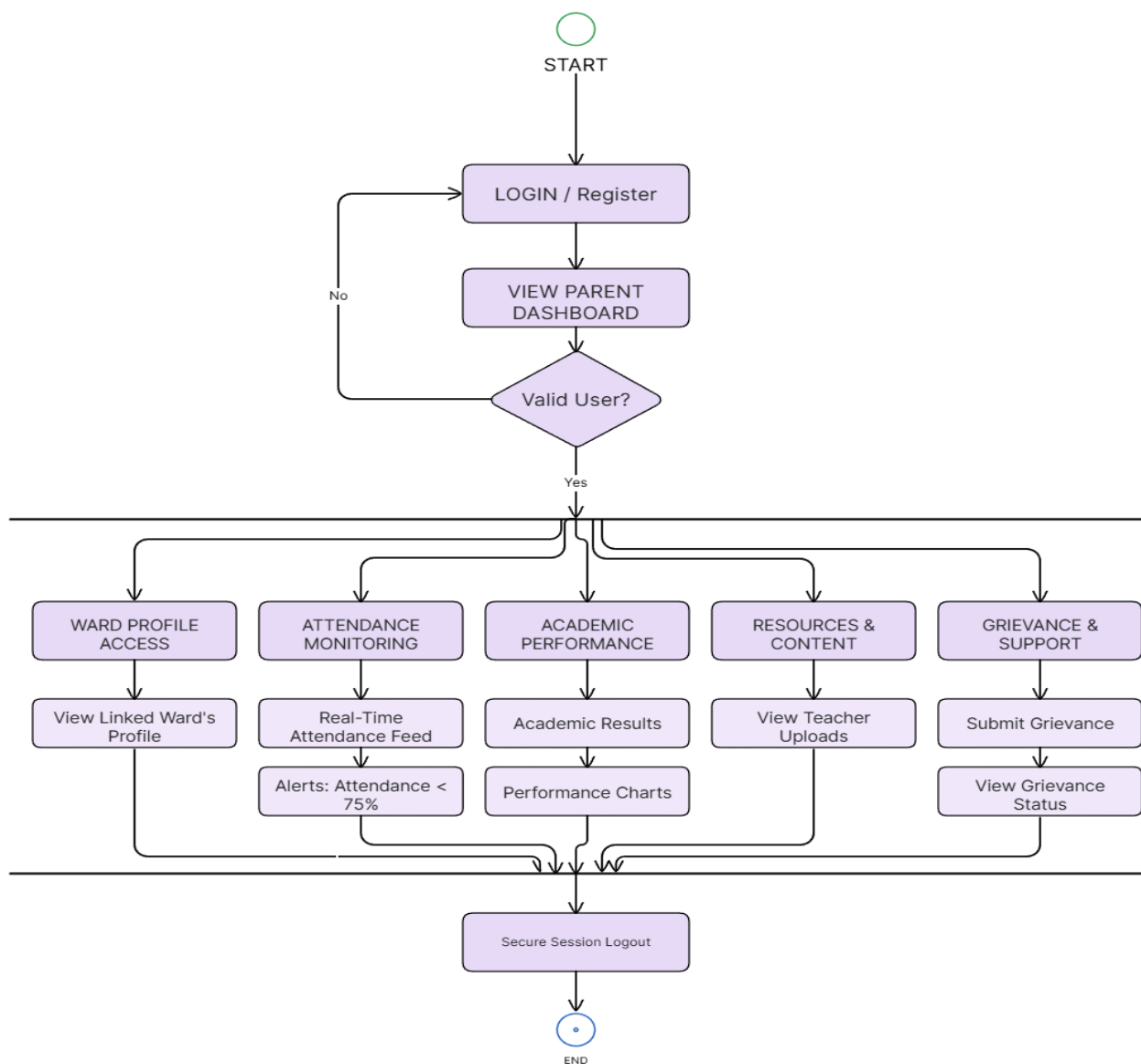


FIGURE -6 ACTIVITY DIAGRAM [parent]

3.7.2.2 ACTIVITY DIAGRAM [student]:-

The Student Activity Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the workflow of student-related activities within the platform. It includes processes such as login, viewing attendance and marks, downloading study materials, submitting grievances and feedback, participating in SkillHub tasks, and receiving notifications.

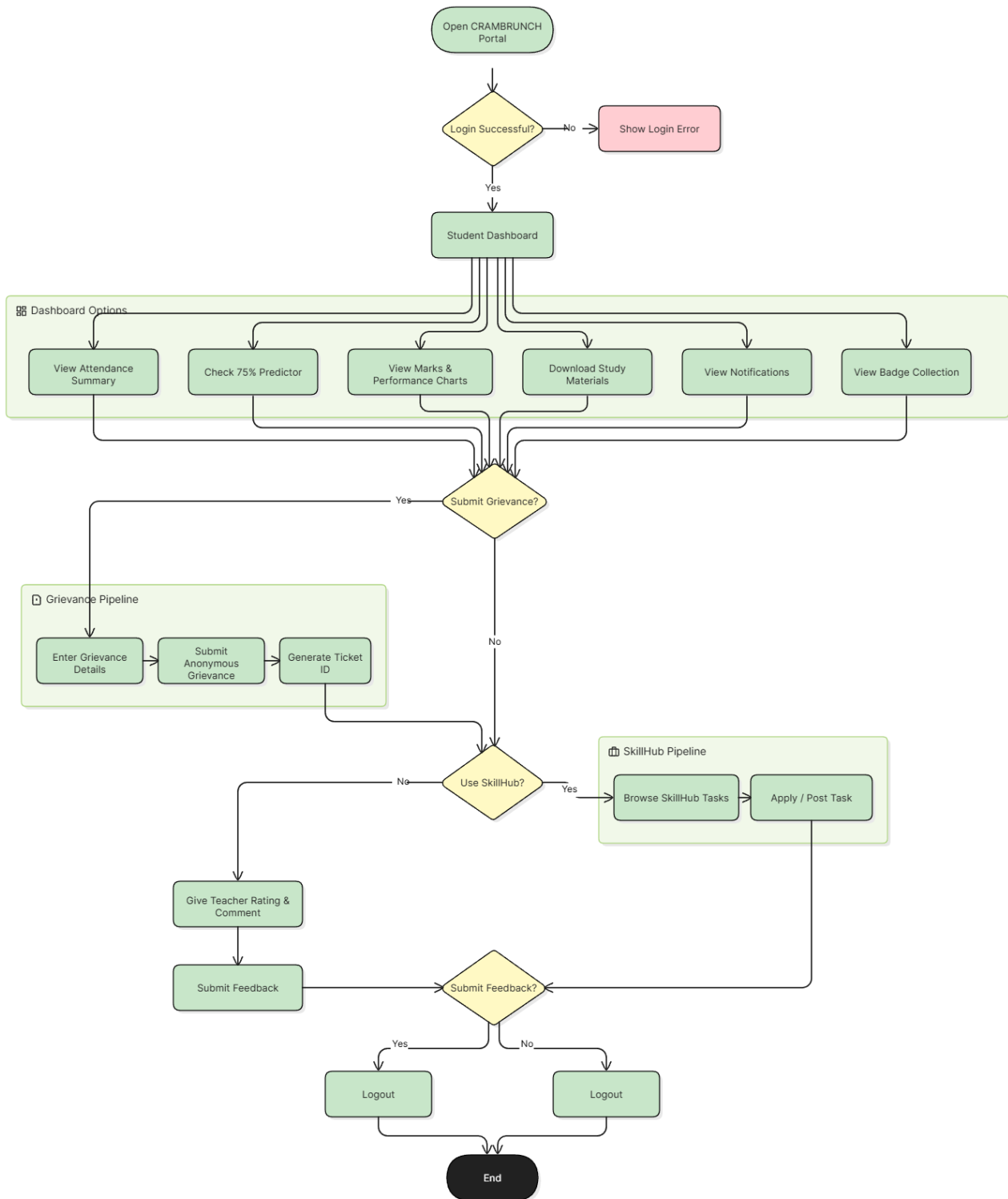


FIGURE -7 ACTIVITY DIAGRAM [student]

3.7.2.3 ACTIVITY DIAGRAM [teacher]:-

The Teacher Activity Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the workflow of teacher-related operations within the platform. It includes login, marking attendance, entering student marks, uploading study materials, viewing timetables, checking feedback ratings, and managing classroom-related activities.

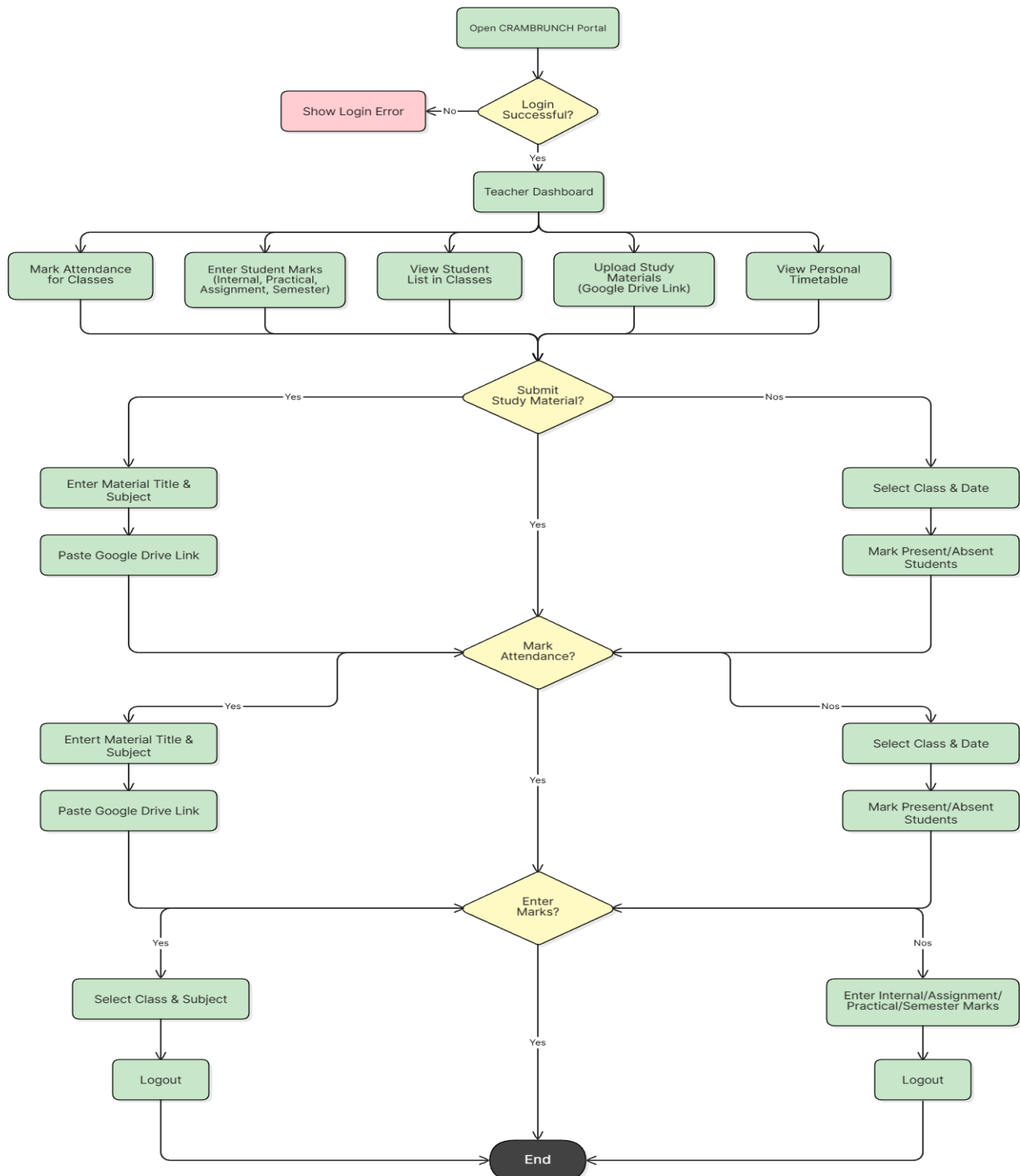


FIGURE -8 ACTIVITY DIAGRAM [teacher]

3.7.2.4 ACTIVITY DIAGRAM [admin]:-

The Admin Activity Diagram of the CRAMBRUNCH Smart Campus Management System represents the sequence of administrative operations performed by the Super Admin. It includes managing users, monitoring system health, handling grievances, managing study materials and SkillHub tasks, awarding badges, and overseeing overall platform activities.

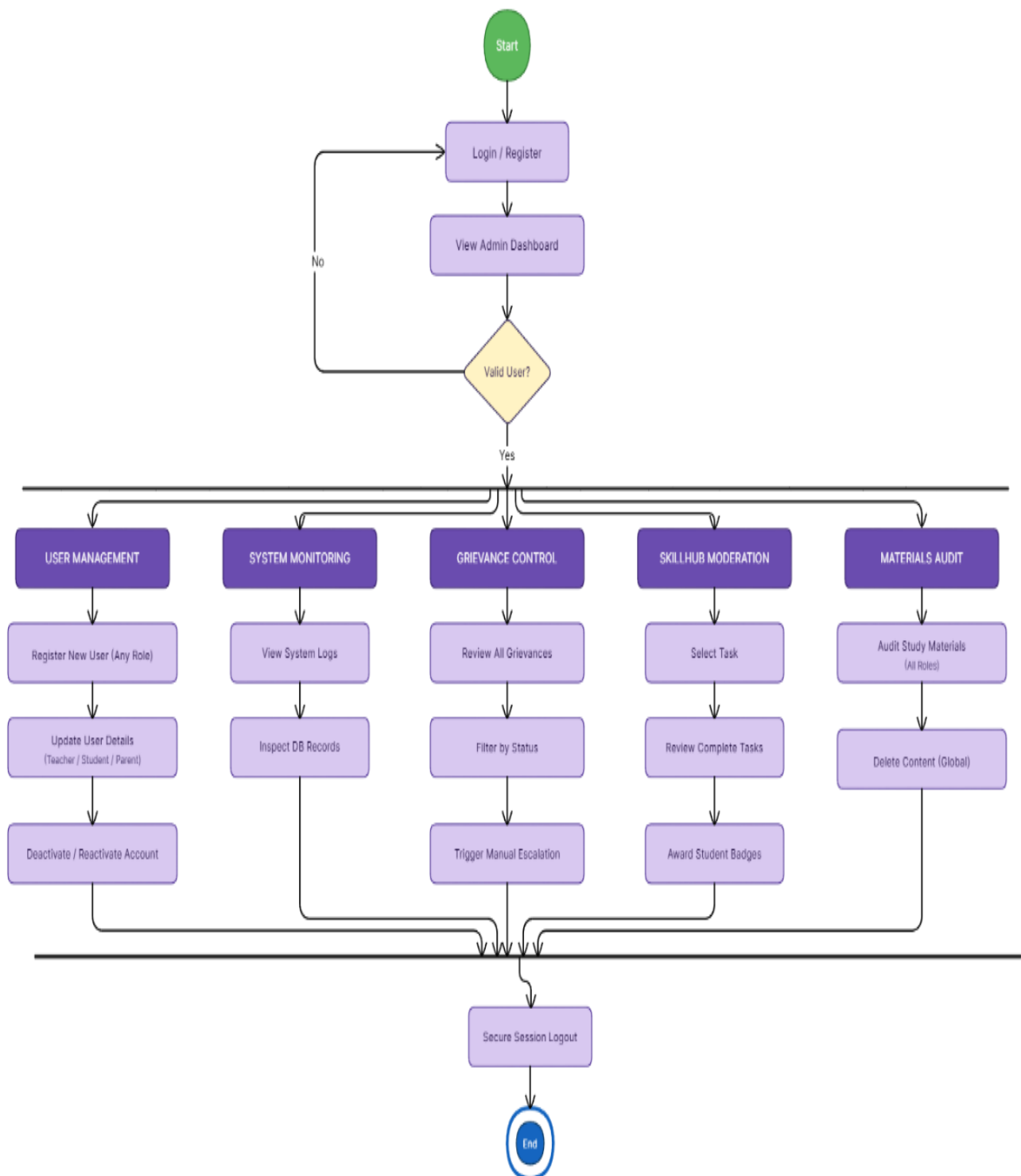


FIGURE -9 ACTIVITY DIAGRAM [admin]

ACTIVITY DIAGRAM [super admin]:-

The Super Admin Activity Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the complete workflow of high-level administrative operations within the platform. It includes registering and managing users, monitoring system performance, handling grievance escalations, managing SkillHub activities, controlling study materials, and maintaining overall system security and functionality.

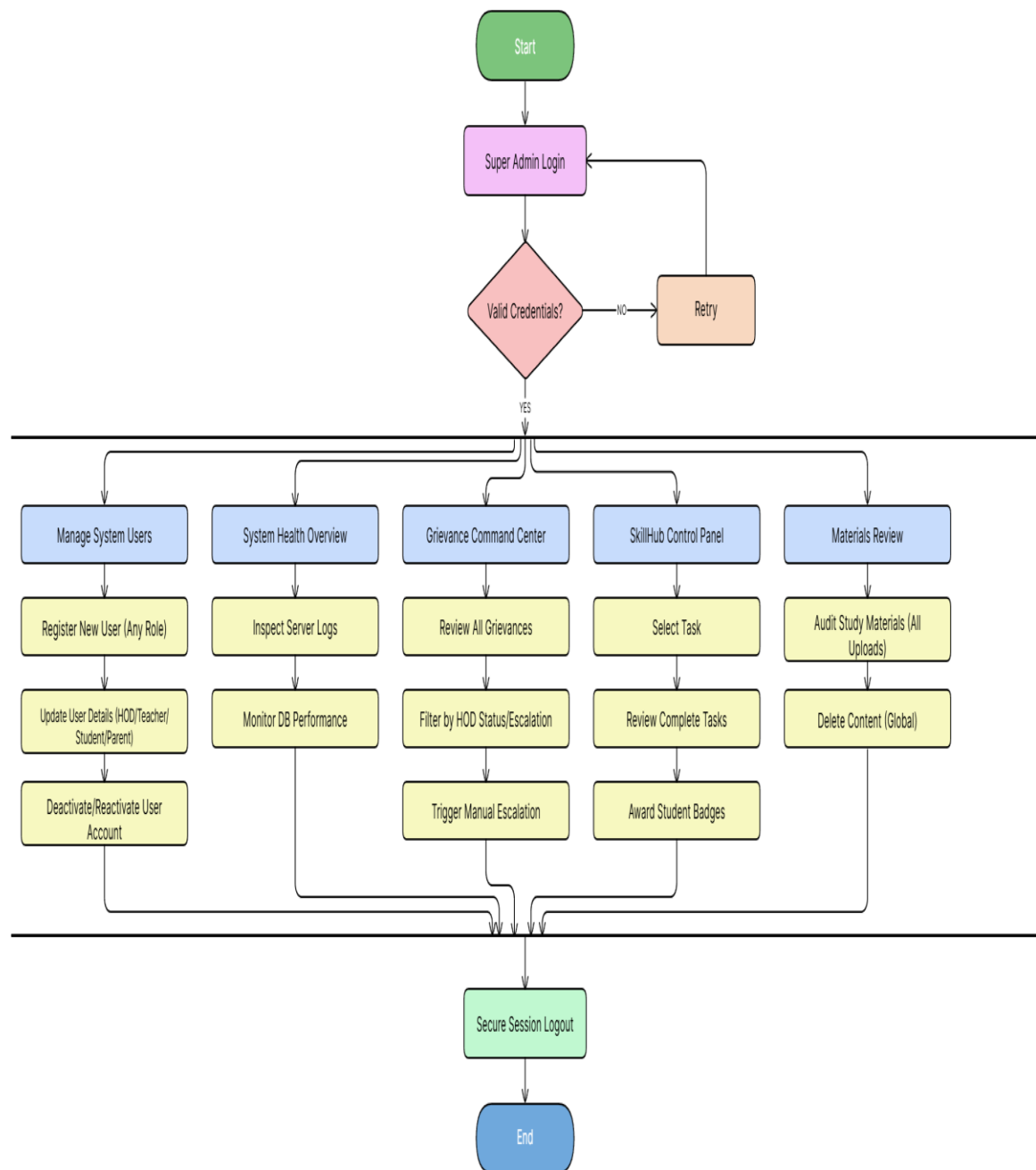


FIGURE -10 ACTIVITY DIAGRAM [super admin]

3.7.3 SEQUENCE DIAGRAM:-

The Sequence Diagram of the CRAMBRUNCH Smart Campus Management System represents the interaction between users, system modules, and the database in a time-based sequence. It illustrates how requests such as login, attendance marking, marks submission, grievance handling, and notifications are processed step by step within the system.

SEQUENCE DIAGRAM [parent]:-

The Parent Sequence Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the sequence of interactions between the parent, system modules, and the database. It shows processes such as parent login, viewing the ward's attendance and marks, receiving attendance alerts, checking notifications, and monitoring academic performance step by step.

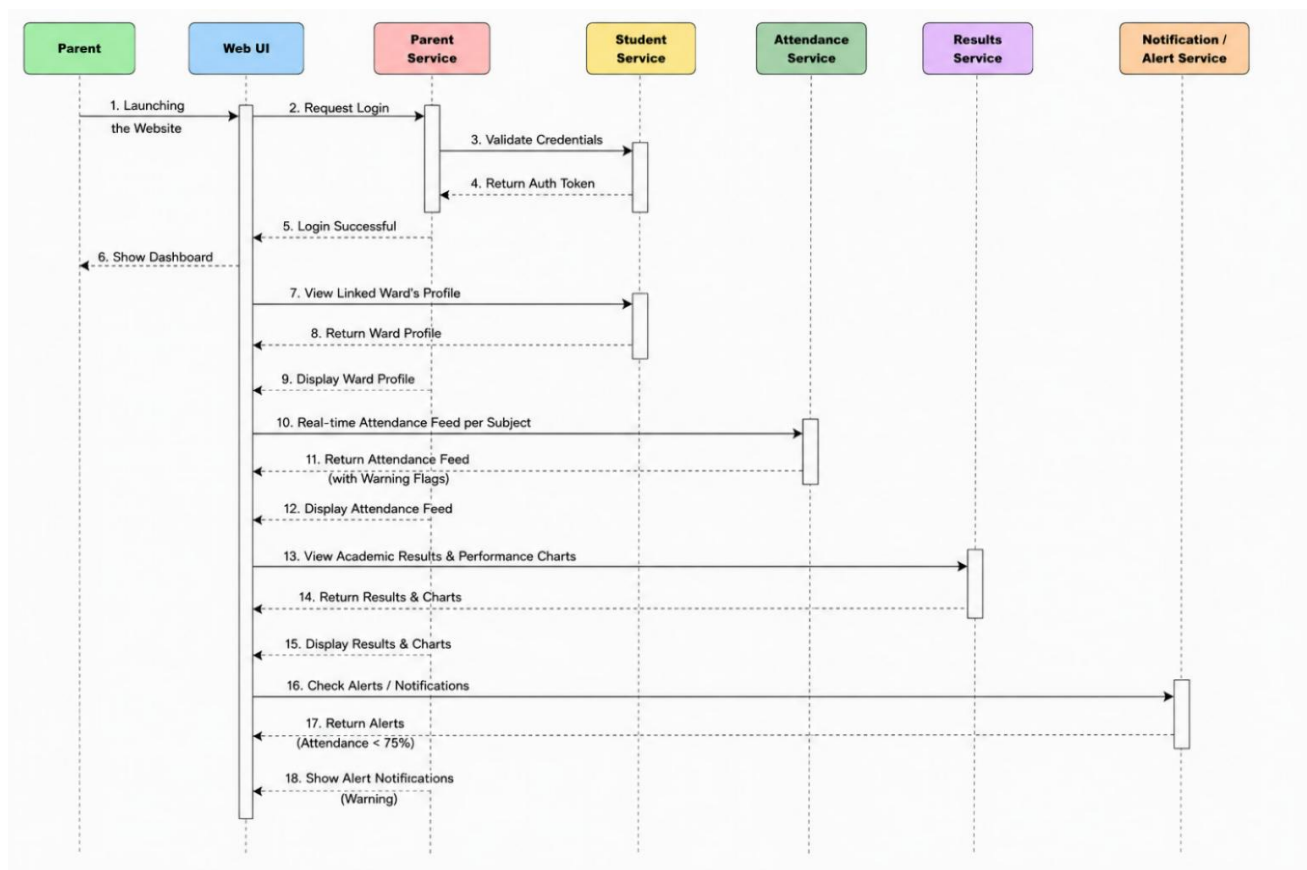


FIGURE -11 SEQUENCE DIAGRAM [parent]

SEQUENCE DIAGRAM [student]:-

The Student Sequence Diagram of the CRAMBRUNCH Smart Campus Management System represents the interaction between the student, system modules, and the database in a sequential manner. It illustrates processes such as student login, viewing attendance and marks, downloading study materials, submitting grievances and feedback, participating in SkillHub tasks, and receiving notifications.

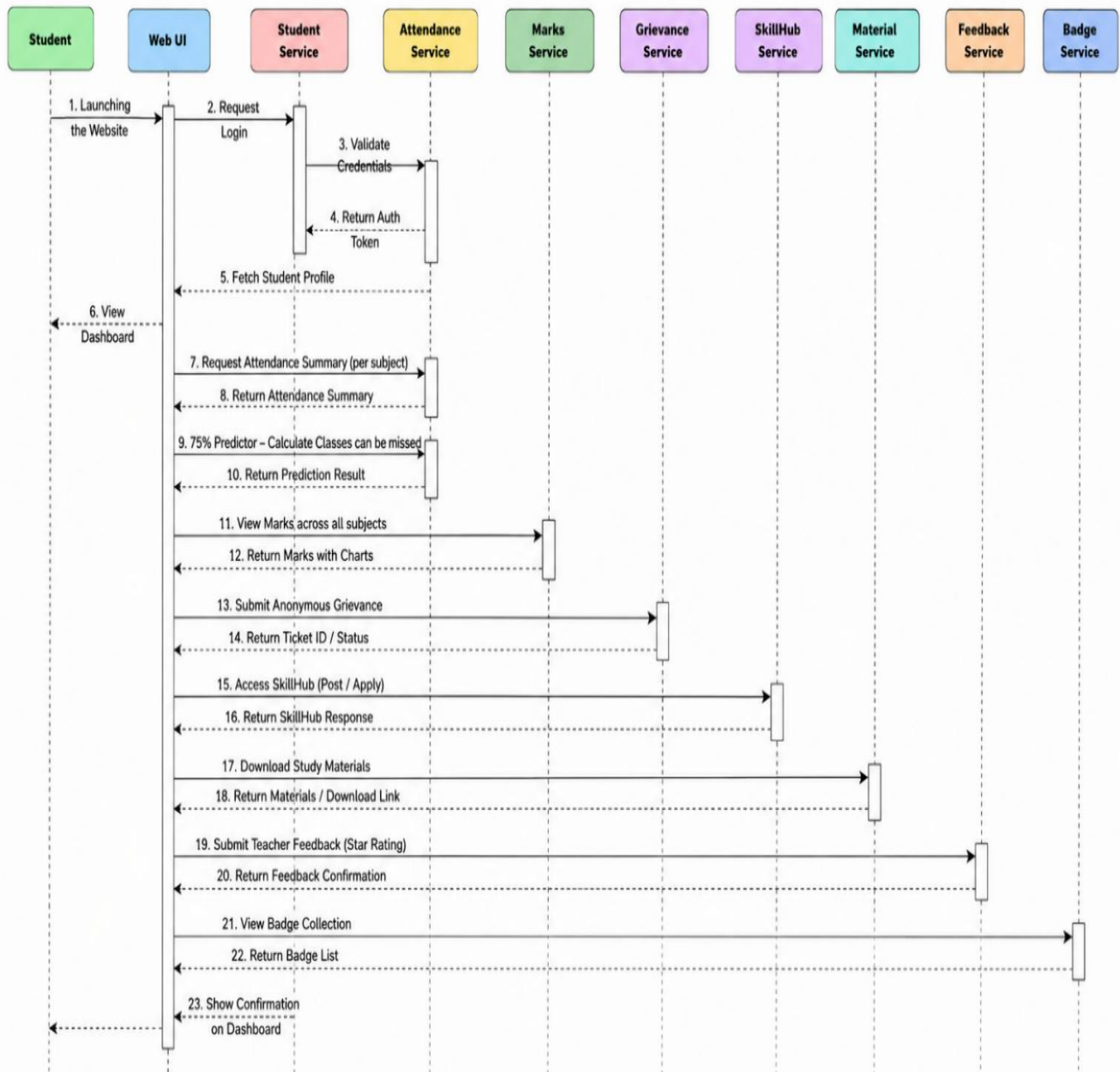


FIGURE -12 SEQUENCE DIAGRAM [student]

SEQUENCE DIAGRAM [teacher]:-

The Teacher Sequence Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the sequence of interactions between the teacher, system modules, and the database. It includes processes such as teacher login, marking attendance, entering marks, uploading study materials, managing timetables, viewing feedback, and sending notifications

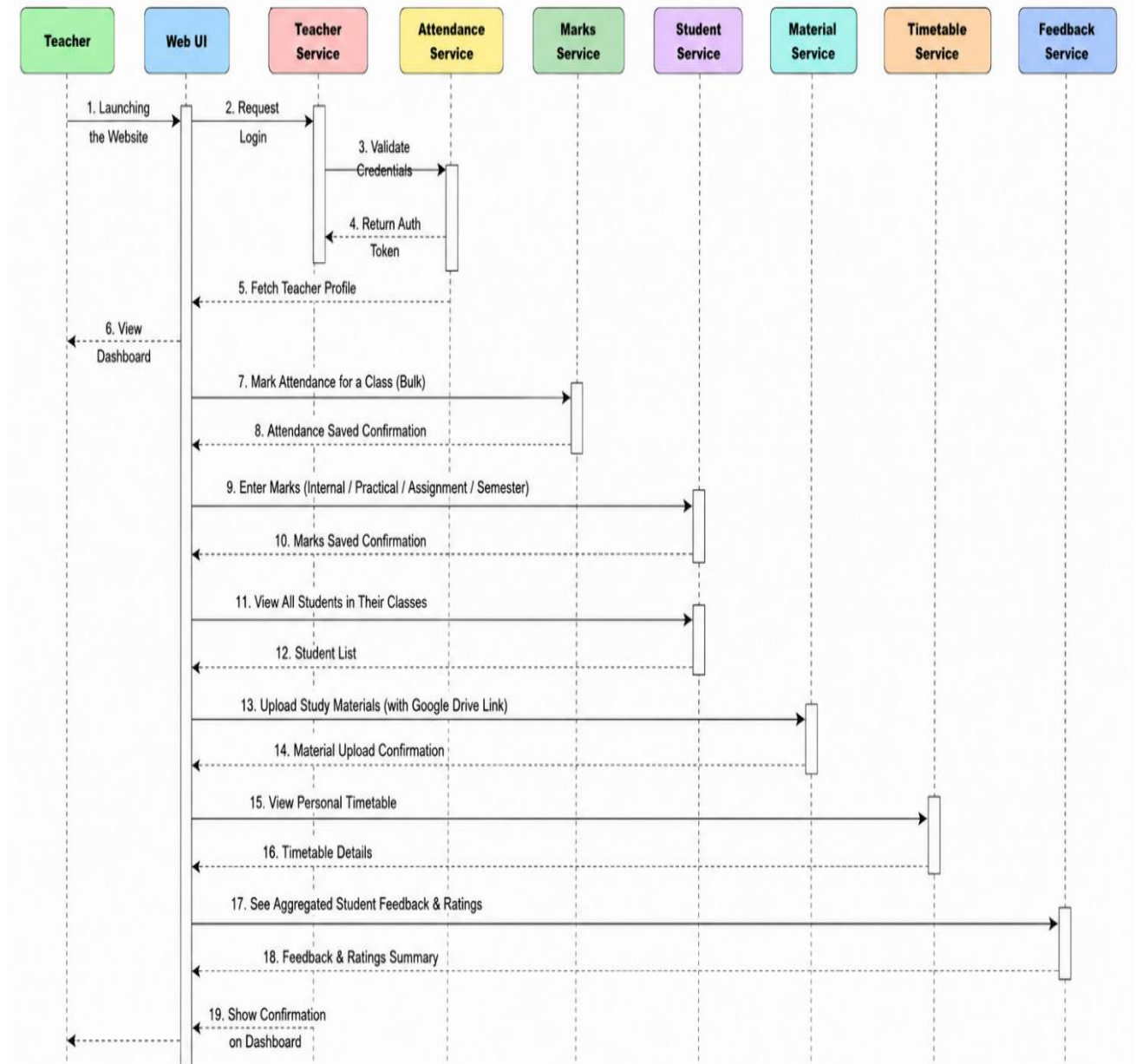


FIGURE -13 SEQUENCE DIAGRAM [teacher]

3.7.5 SEQUENCE DIAGRAM [admin] :-

The Admin Sequence Diagram of the CRAMBRUNCH Smart Campus Management System represents the sequence of interactions between the admin, system modules, and the database. It illustrates activities such as admin login, user management, grievance handling, monitoring system health, managing SkillHub tasks, controlling study materials, and generating reports step by step.

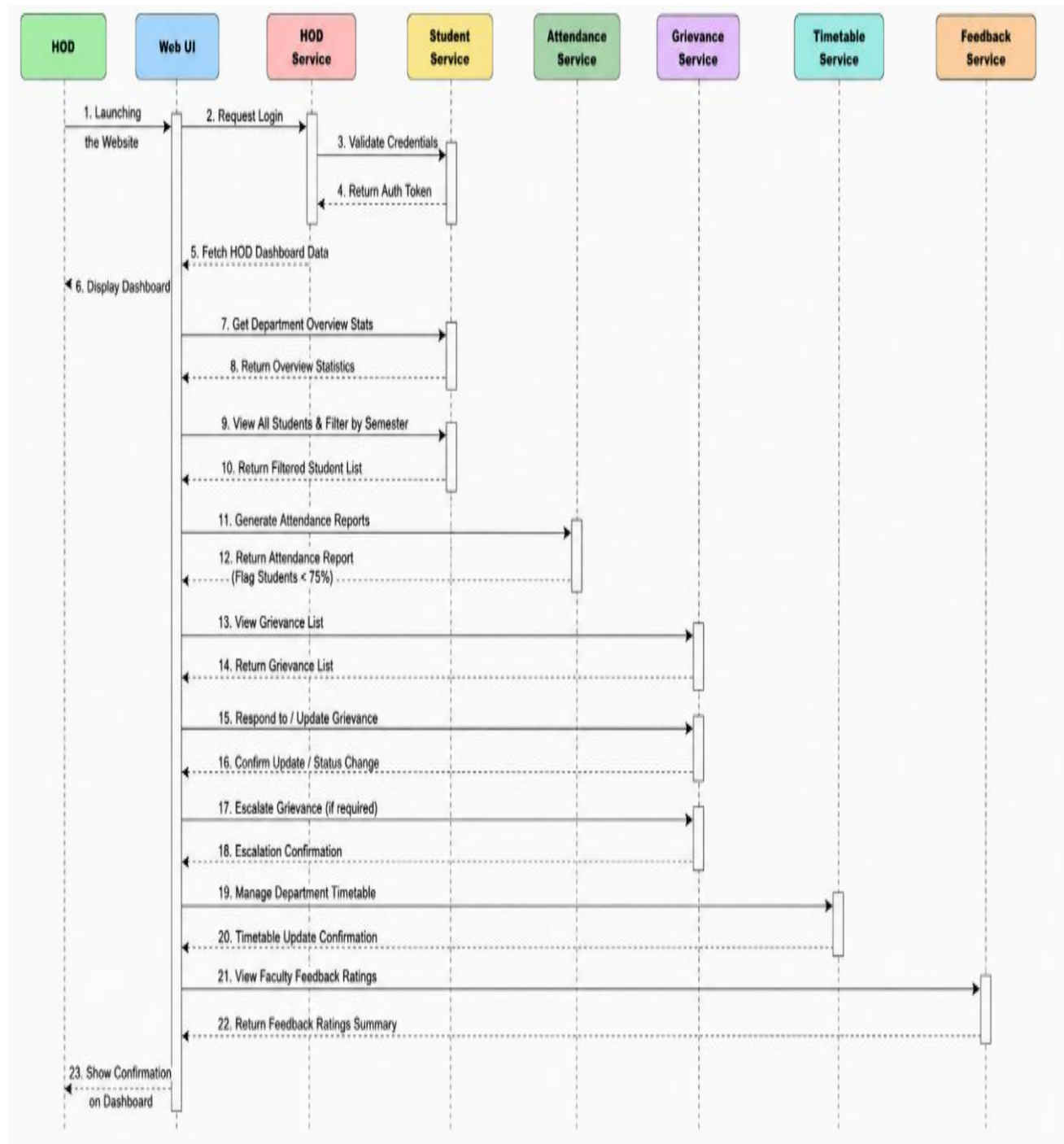


FIGURE -14 SEQUENCE DIAGRAM [admin]

SEQUENCE DIAGRAM [super admin]:-

The Super Admin Sequence Diagram of the CRAMBRUNCH Smart Campus Management System illustrates the interaction flow between the Super Admin, system modules, and the database in a sequential order. It includes processes such as secure login, managing all user roles, monitoring system activities, handling grievance escalations, managing SkillHub and study materials, and maintaining overall system control and security.

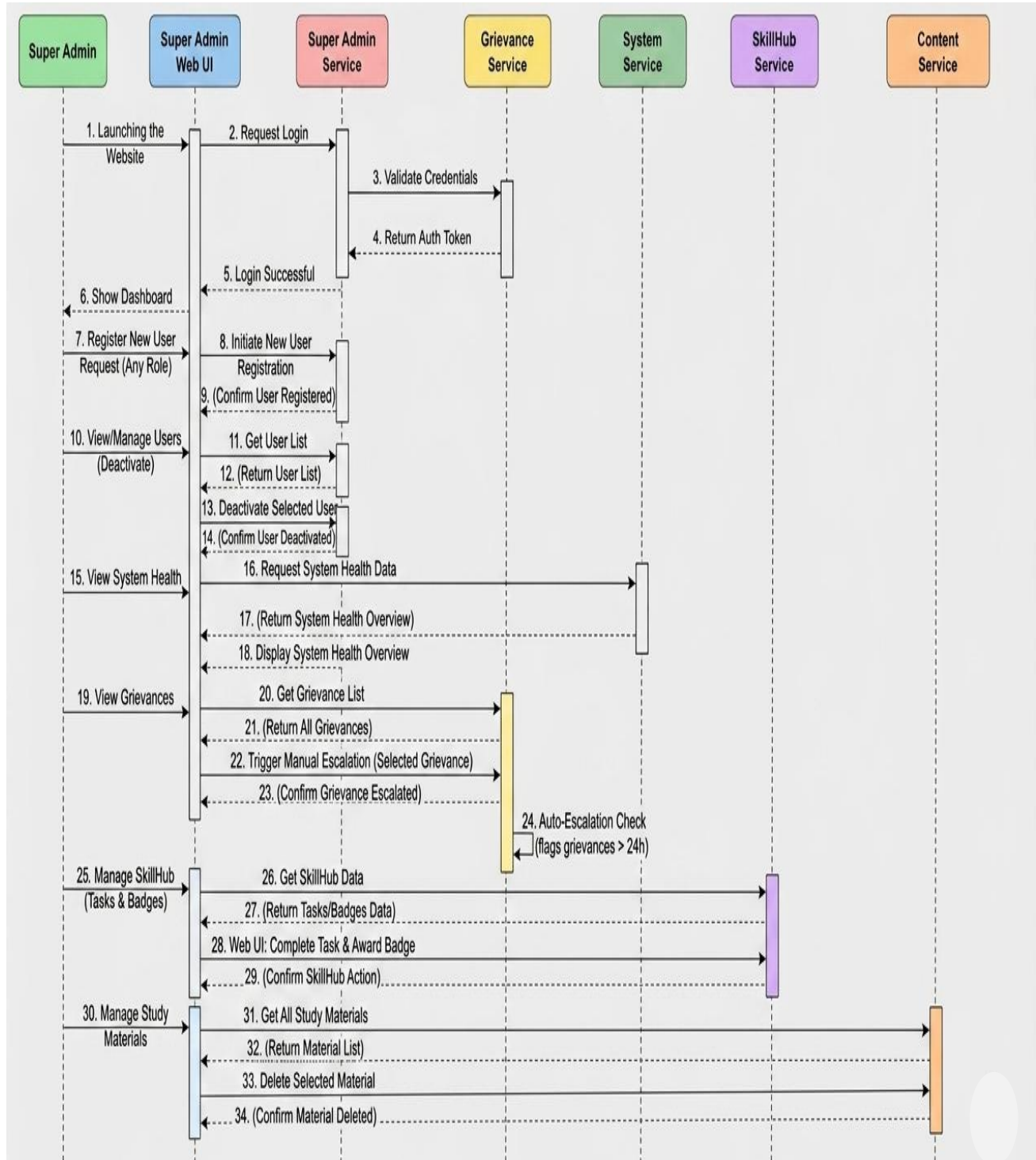


FIGURE -14 SEQUENCE DIAGRAM [super admin]

CHAPTER-4

(TESTING PROCESSES)

TESTING PROCESSES:

Testing ensures CRAMBRUNCH works correctly across all modules like attendance and grievances. It finds and fixes errors before real use, matching the manual testing done in the project.

4.1 Testing Objectives

Testing verifies system reliability and user needs. It checks if features like one-tap attendance or skill badges work as planned. Overall, it improves speed and catches issues early.

4.2 Types of Testing

CRAMBRUNCH used these standard methods step-by-step.

- Unit Testing: Tests single parts like login forms or grievance submission.
- Integration Testing: Checks module links, e.g., attendance updates parent feeds.
- System Testing: Runs full app end-to-end for all roles.

4.2.1 Unit Testing:-

Unit Testing

Unit testing checks the smallest parts of CRAMBRUNCH, such as a login function, attendance calculation, or grievance submission, to make sure each part works correctly on its own. It is usually done by developers during the coding stage so that bugs can be found early and fixed quickly.

Purpose

The main purpose of unit testing is to verify that each function gives the correct output for a given input. For CRAMBRUNCH, this could mean checking whether a student login works, whether attendance is saved properly, or whether the 75% predictor gives the right result.

How it is done

Unit tests are written for individual functions and run separately from the full system. Developers usually test valid inputs, invalid inputs, and edge cases to make sure the code behaves correctly in all situations.

Example for CRAMBRUNCH

- Test the login function with correct email and password.
- Test the attendance function with a valid class and student list.
- Test the grievance form with an empty message to check error handling.
- Test the attendance predictor with different present and absent counts.

Outcome

If the unit test passes, the function is working properly. If it fails, the developer can fix that specific part before testing the whole system. This makes CRAMBRUNCH more reliable and easier to maintain.

4.2.2 Integration Testing

Integration testing checks whether different modules of CRAMBRUNCH work properly together after unit testing is completed. It focuses on the interaction between modules, such as how attendance updates are linked with parent notifications or how grievances move to HOD escalation.

Purpose

The main purpose of integration testing is to make sure data flows correctly from one module to another. It helps find problems in communication between components before the full system is tested.

How it is done

In CRAMBRUNCH, modules are connected and tested in groups. For example, the teacher attendance module can be tested together with the notification module to see whether alerts are sent correctly to students or parents.

Example for CRAMBRUNCH

- Test whether marking attendance updates the student record correctly.
- Test whether a grievance submitted by a student appears in the HOD dashboard.
- Test whether completed SkillHub tasks update the badge system.
- Test whether notifications are generated after marks or attendance changes.

Outcome

If integration testing passes, it means the modules are working well together and the system is ready for larger-level testing. This ensures CRAMBRUNCH can handle real workflows without breaking data connections.

Security Testing

Security testing ensures that CRAMBRUNCH is protected from unauthorized access, data leaks, and other security issues. It checks whether user login, role-based access, and stored data are safe from misuse.

- **Authentication Testing:** This checks whether only valid users can log in to the system. It also verifies that usernames and passwords are handled securely.
- **Authorization Testing:** This ensures that each user can access only the features meant for their role. For example, a student should not be able to view admin data.
- **Data Protection Testing:** This checks whether important data such as passwords, attendance, grievances, and marks are stored securely.
- **Session Security Testing:** This verifies that user sessions are properly managed and that unused sessions do not remain open.
- **Input Validation Testing:** This checks whether the system blocks invalid or harmful input to prevent attacks like injection or form misuse.
- **Role-Based Access Testing:** This makes sure that students, teachers, parents, HODs, and admins can only use their own dashboards and functions.
- **Audit Log Testing:** This checks whether the system records important actions like login, data updates, and grievance changes for future tracking.

Security testing is important because CRAMBRUNCH handles sensitive academic and personal information. A secure system helps build trust and keeps user data safe.

CHAPTER-5

(RESULTS AND DISCUSSIONS)

5.RESULT AND DISCUSION :-

The CRAMBRUNCH platform was successfully developed as a web-based college management system that combines attendance, grievance handling, skill exchange, and communication features in one place. It reduced the need for manual work and provided a simple and organized way for students, teachers, parents, and admins to interact with the system.

❖ Result:-

The final system works smoothly and supports all major planned modules. Users can log in based on their role and access only the features meant for them. Teachers can mark attendance, students can submit complaints, parents can view updates, and admins can manage reports and records easily. The system also gives quick notifications and keeps data properly stored in the database.

❖ Discussion:-

The main advantage of CRAMBRUNCH is that it brings all campus activities into one digital platform. It improves communication, saves time, and reduces errors caused by paper-based work. The skill exchange and micro-internship features make it more useful than a normal management system because they help students grow beyond academics.

The testing results showed that the system is stable, easy to use, and suitable for daily college use. Some minor issues such as UI adjustments and response time improvements were handled during testing. Overall, the project meets its objectives and shows that a smart, role-based college management system can improve campus efficiency and transparency.

SNAPSHOTS

Landing Page

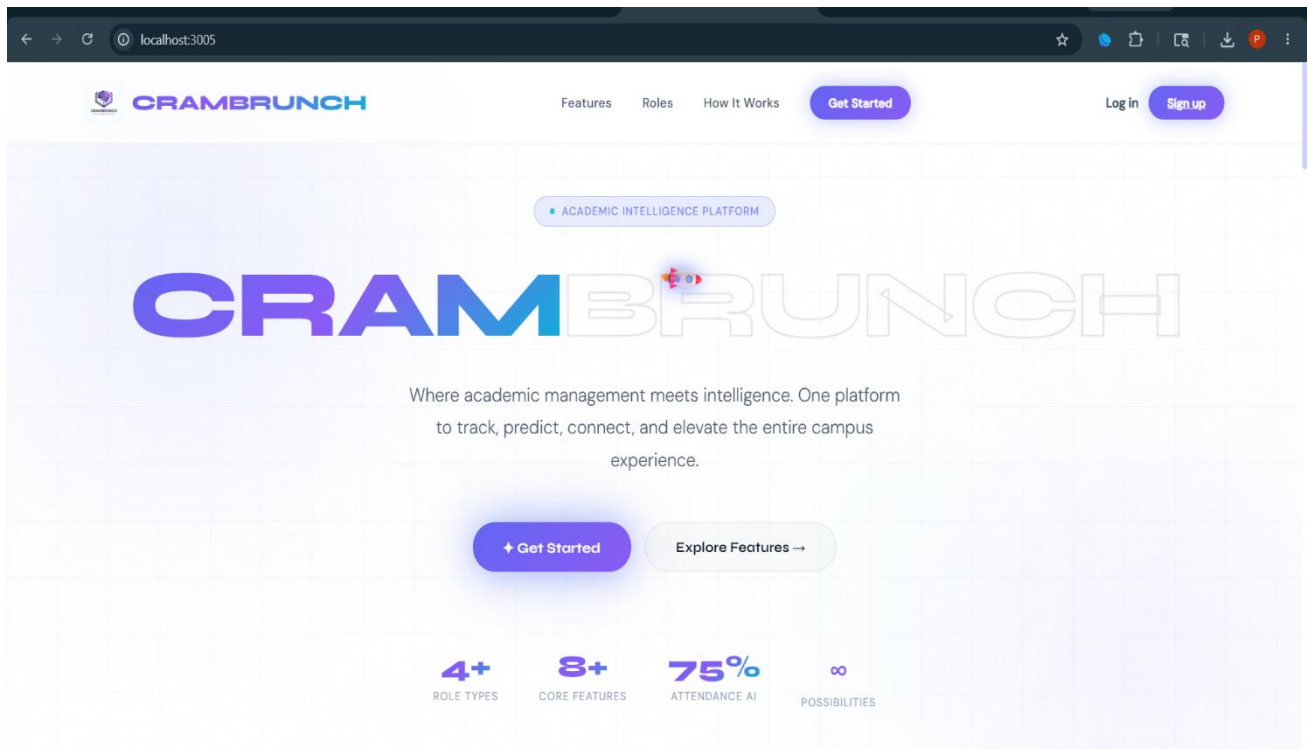


FIGURE -15 Landing Page

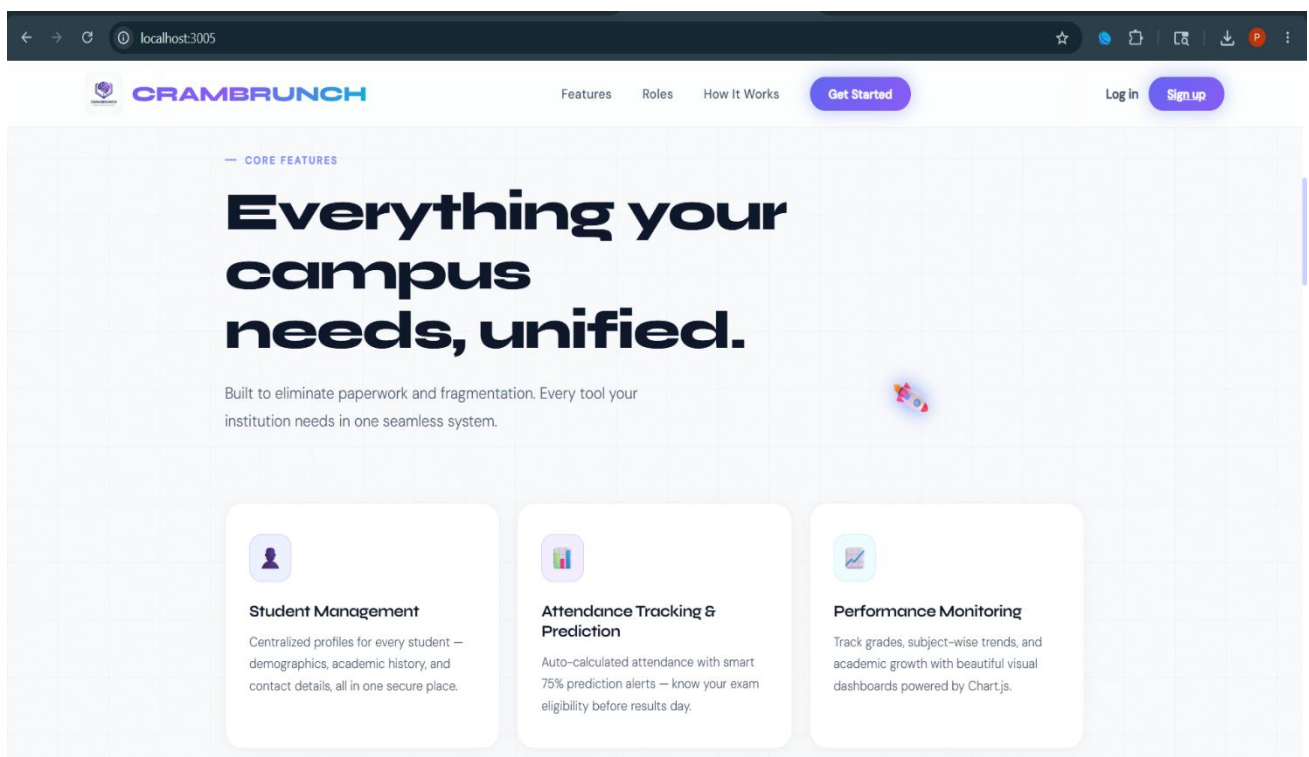


FIGURE -16 Landing Page

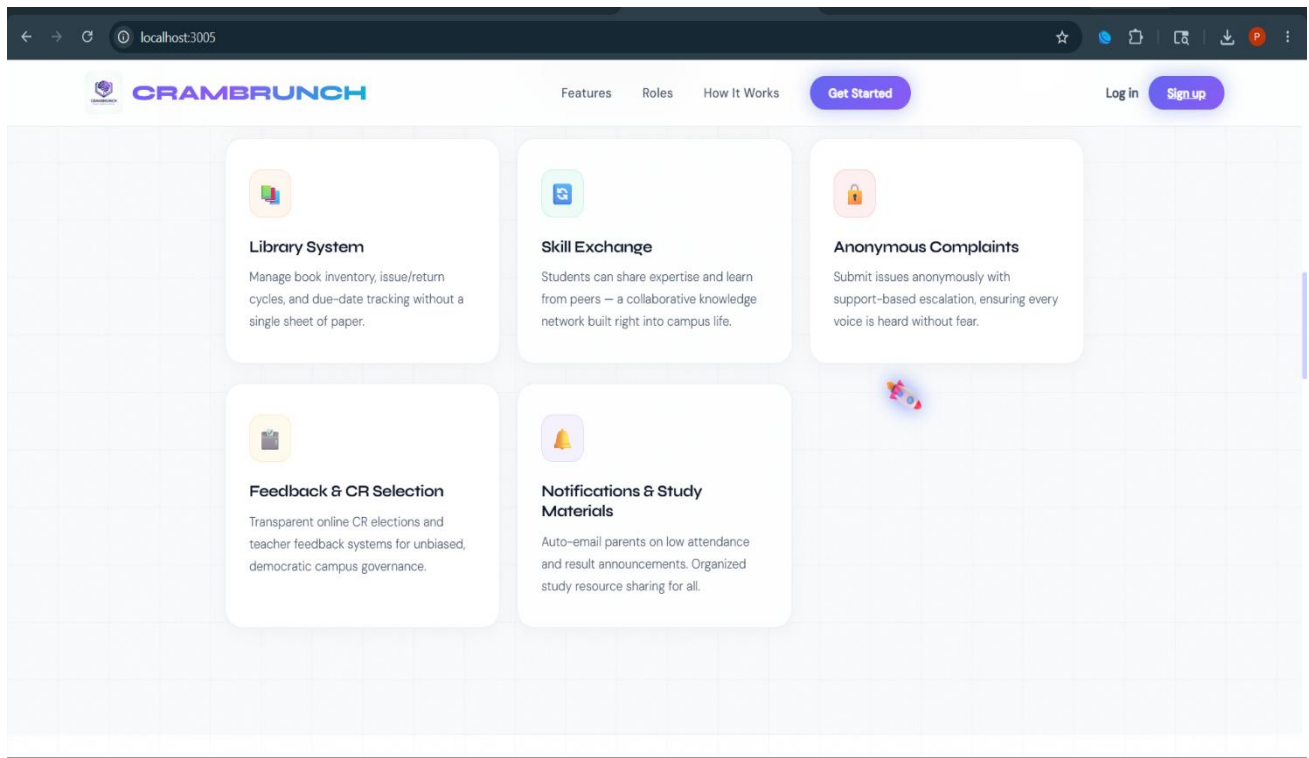


FIGURE -17 Landing Page

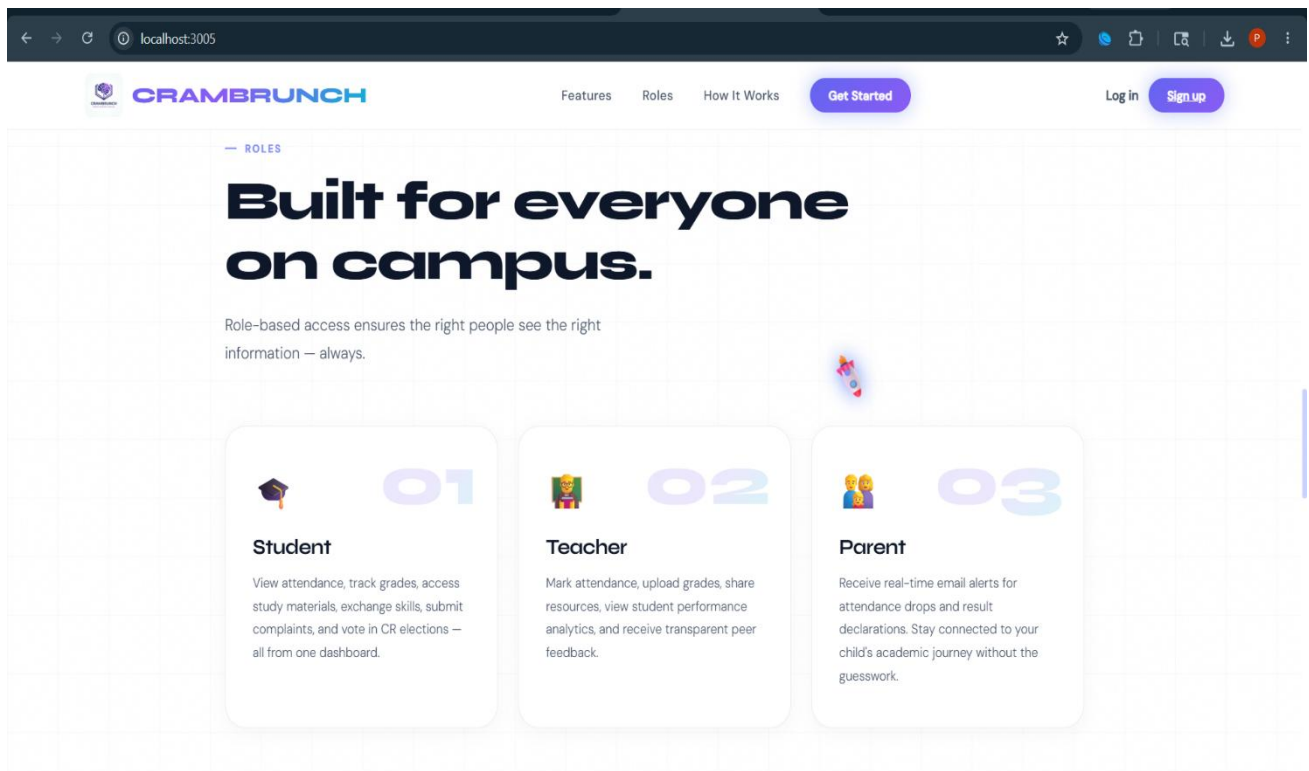


FIGURE -18 Landing Page

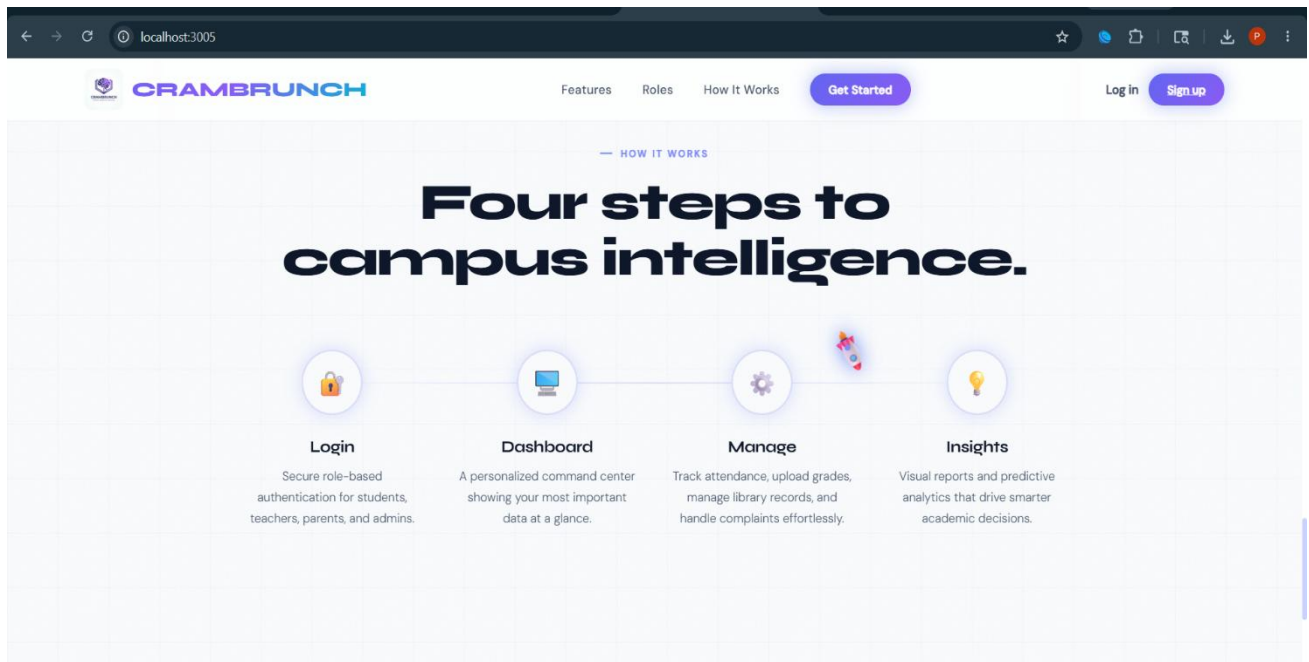


FIGURE -19 Landing Page

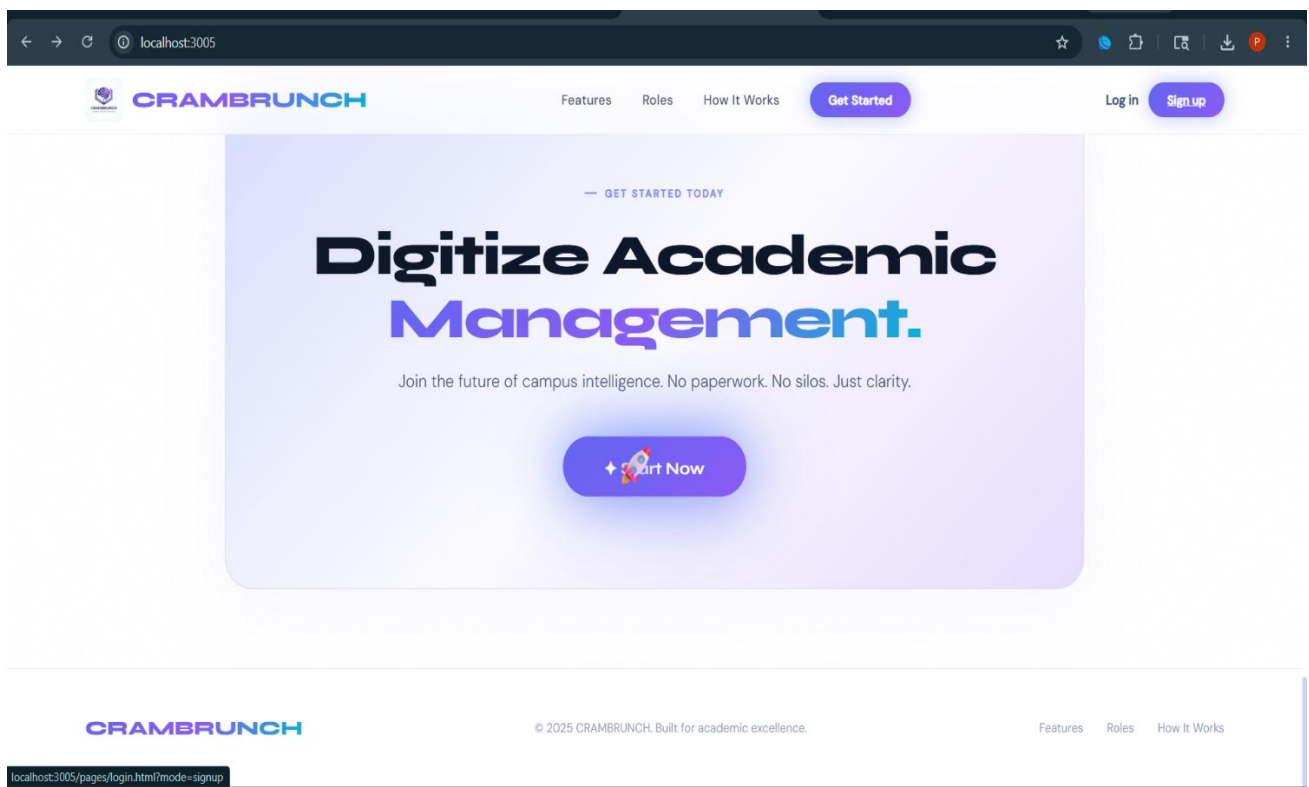


FIGURE -20 Landing Page

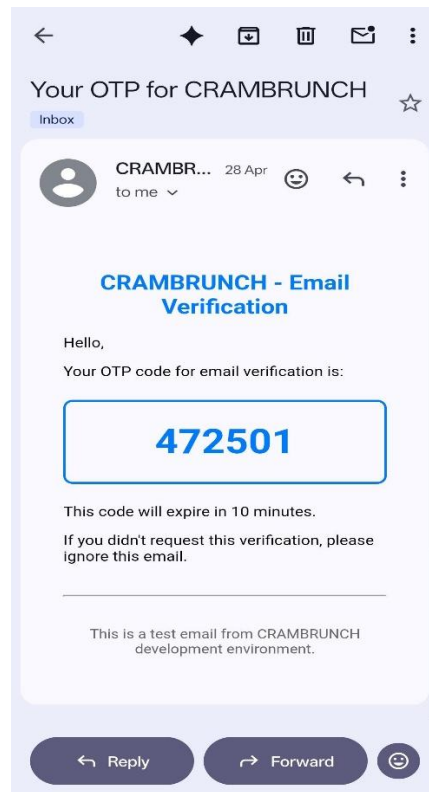


FIGURE -22 (otp verification)

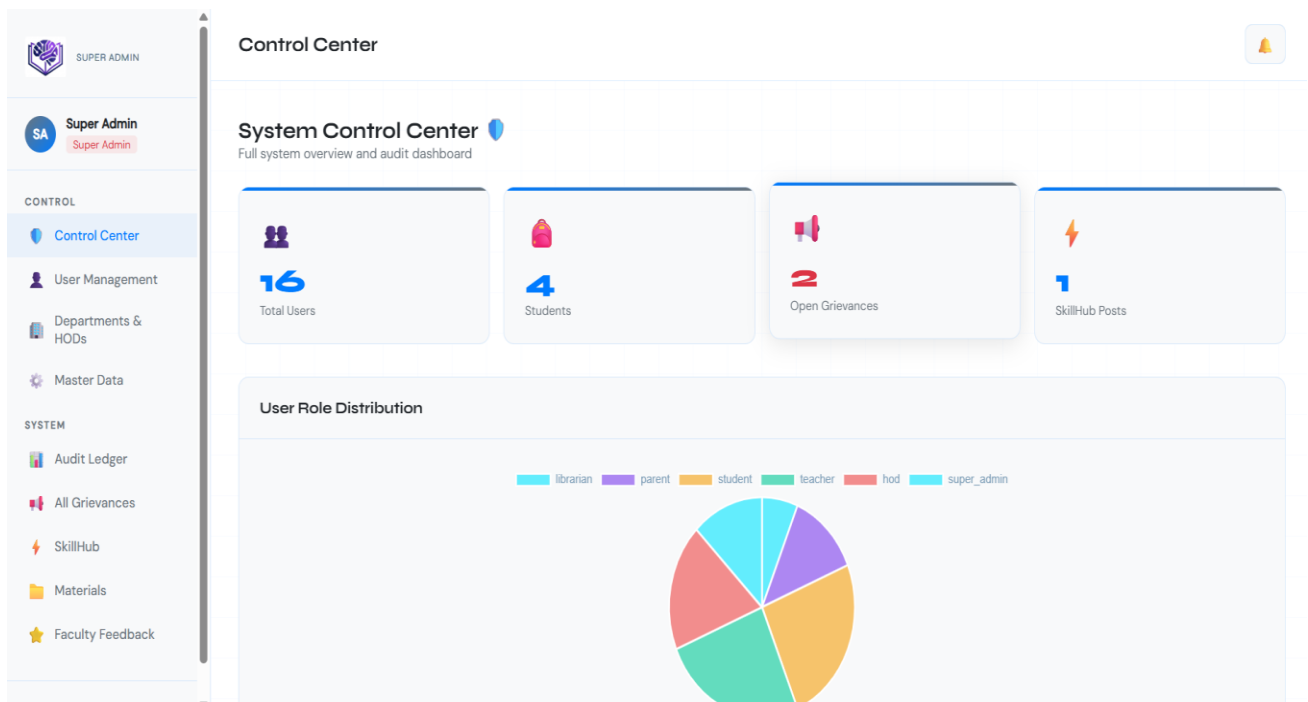


FIGURE -23 Super Admin Dashboard

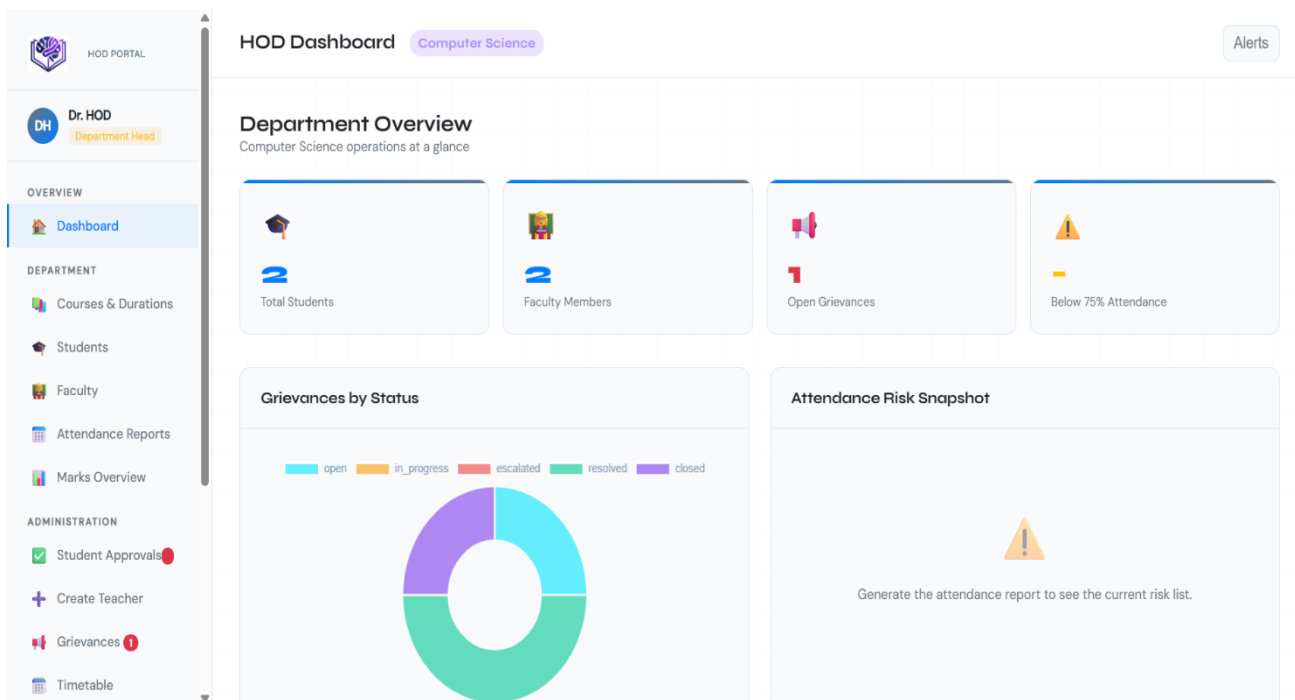


FIGURE -24 HOD Dashboard

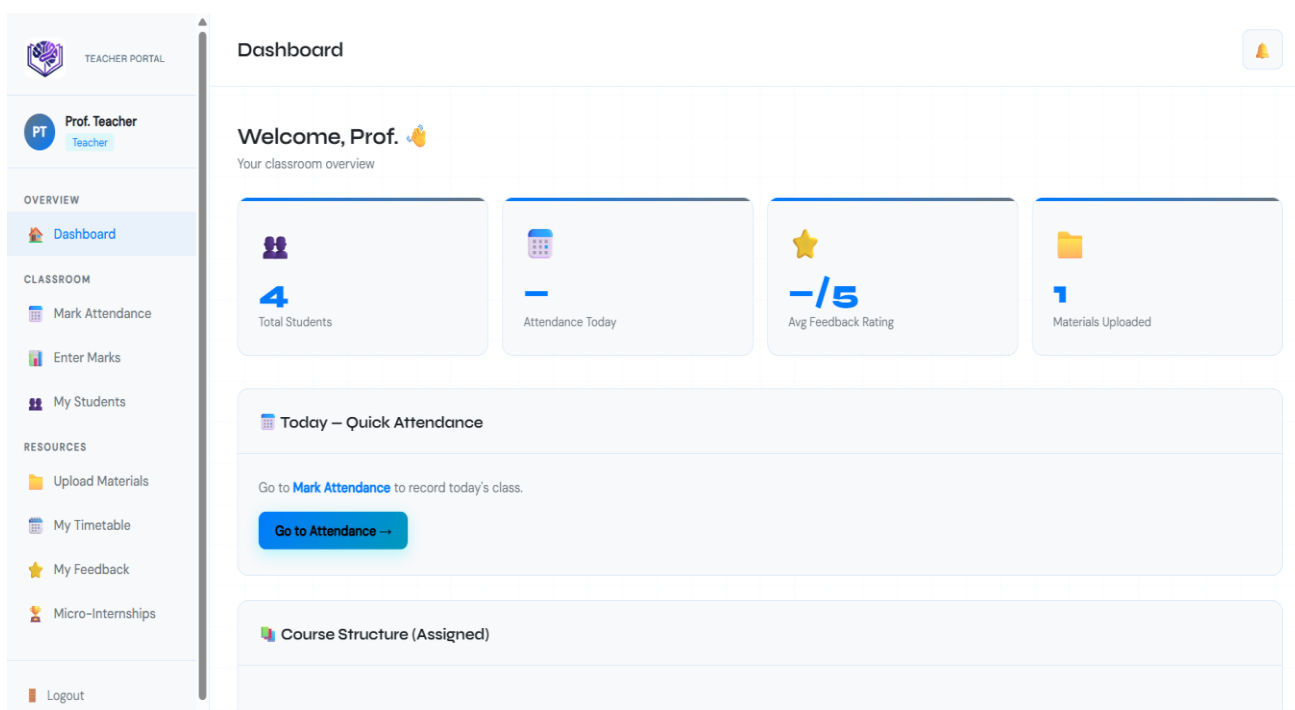


FIGURE -25 Teacher Dashboard

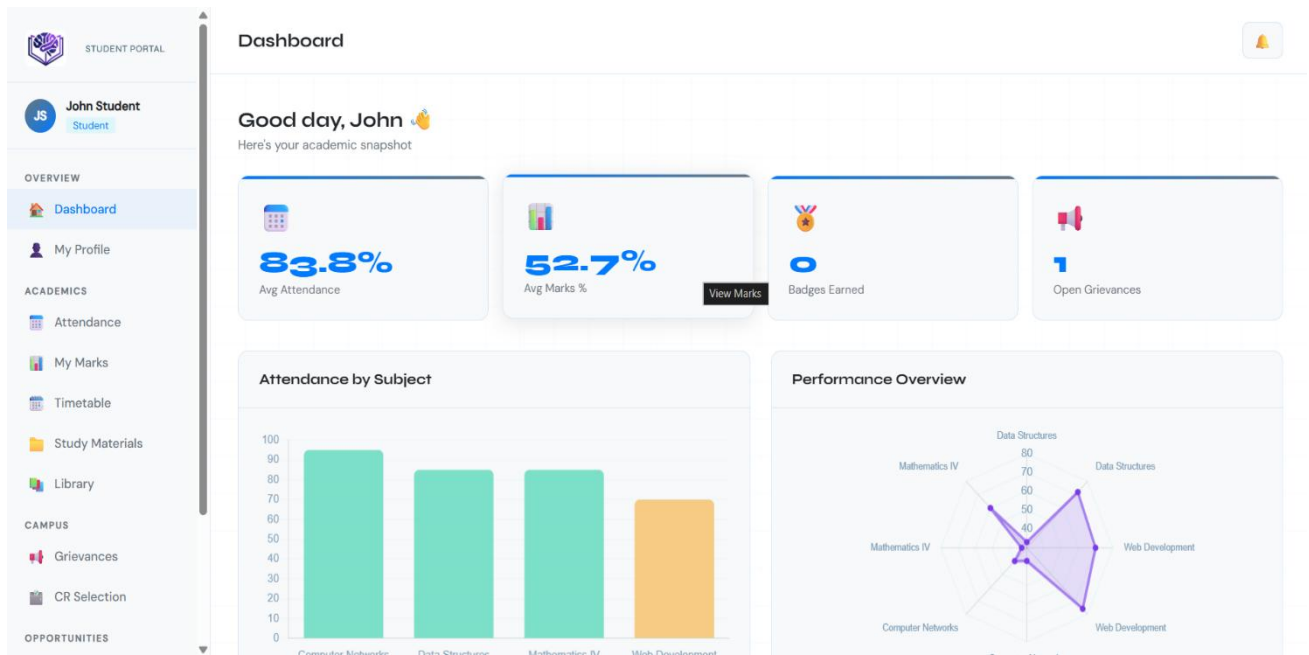


FIGURE -26 Student Dashboard

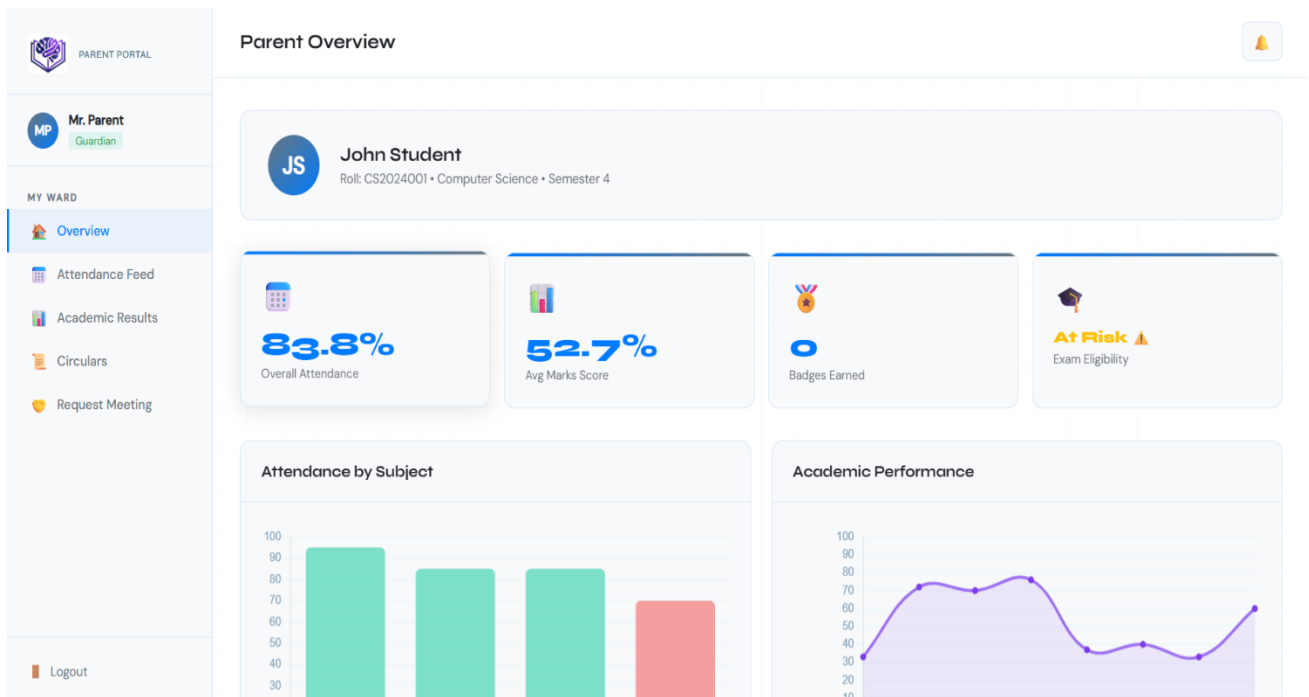


FIGURE -27 Parent Dashboard

CHAPTER-6

(FUTURE SCOPE AND CONCLUSION)

6.1 Future Scope

The future scope of CRAMBRUNCH is very wide because it can be improved with new technologies and additional modules. The system can later include AI-based suggestions, cloud storage, mobile app support, and better analytics to make college management smarter and faster.

- **AI Features:** The system can suggest useful actions automatically, such as predicting attendance issues or recommending tasks for students. This will make the platform more intelligent and helpful.
- **Cloud Deployment:** CRAMBRUNCH can be moved from localhost to cloud servers so that users can access it from anywhere. This will also improve storage, backup, and accessibility.
- **Mobile Application:** A dedicated mobile app can be developed for easier access by students, parents, and teachers. This will make the system more convenient for daily use.
- **Advanced Analytics:** The system can include performance charts and reports to help colleges understand student progress better. This will support better decision-making.
- **More Modules:** Future versions can add online exams, fee payment, hostel management, and library automation. This will make CRAMBRUNCH a complete campus management solution.
- **Chatbot Support:** A smart chatbot can be added to answer common questions instantly. This can reduce manual support work and improve user experience.

Overall, CRAMBRUNCH has strong future scope because it can grow into a full digital campus platform with smart, secure, and user-friendly features.

6.2 Conclusion

CRAMBRUNCH is a modern and efficient college management platform designed to simplify academic and campus-related tasks. It provides a single digital system where students, teachers, parents, HODs, and admins can work together in a more organized way. By replacing manual work with digital processes, the system saves time, reduces errors, and improves overall management.

The project includes important features such as attendance marking, attendance prediction, grievance submission, skill exchange, study material sharing, feedback collection, and notification alerts. These features make the system more useful than a basic student management application because it supports both academic management and student development. It also helps parents stay informed about their ward's progress and attendance.

The system is built using a simple and flexible technology stack, which makes it easy to maintain and expand in the future. Its role-based access control ensures that every user sees only the information and features meant for them. This improves security and also makes the system easier to use.

In conclusion, CRAMBRUNCH is a complete and practical solution for modern educational institutions. It improves communication, transparency, and efficiency across the campus. With future enhancements, it can become even more powerful and can support a wider range of college management activities.

➤ **References / Existing System Study**

1. Google Classroom – <https://classroom.google.com>
Studied for assignment management and classroom organization.
2. Moodle LMS – <https://moodle.org>
Referred for learning management and academic tracking modules.
3. ERPNext Education Module – <https://erpnext.com/education>
Used as reference for attendance and administration workflows.
4. Microsoft Teams for Education – <https://www.microsoft.com/en-in/education/products/teams>
Studied for communication and notification systems.
5. Urban Company – <https://www.urbancompany.com>
Inspired the SkillHub and service exchange concept.
6. Coursera – <https://www.coursera.org>
Referred for badge-based learning and skill verification.
7. LinkedIn Learning – <https://www.linkedin.com/learning>
Inspiration for skill showcase and portfolio features.
8. Research Papers & Articles
 - Smart Campus Management Systems
 - Attendance Monitoring Systems
 - Role-Based Access Control (RBAC)
 - Complaint & Feedback Management Systems
 - Student Skill Development Platforms